TEXAS INSTRUMENTS

OPA171-Q1, OPA2171-Q1, OPA4171-Q1

SBOS556C – JUNE 2011 – REVISED DECEMBER 2015

OPAx171-Q1 36-V, Single-Supply, General-Purpose Operational Amplifier

1 Features

- Qualified for Automotive Applications
- AEC-Q100 Test Guidance With the Following:
  - Device Temperature Grade 1: –40°C to +125°C Ambient Operating Temperature
  - Device HBM ESD Classification Level:
    - Level 3A for OPA171-Q1
    - Level 3A for OPA2171-Q1
    - Level 2 OPA4171-Q1
  - Device CDM ESD Classification Level
    - Level C4A for OPA171-Q1
    - Level C6 for OPA2171-Q1
    - Level C6 for OPA4171-Q1
- Supply Range: 2.7 to 36 V, ±1.35 to ±18 V
- Low Noise: 14 nV/√Hz
- Low Offset Drift: ±0.3 µV/°C (typ)
- RFI Filtered Inputs
- Input Range Includes the Negative Supply
- Input Range Operates to Positive Supply
- Rail-to-Rail Output
- Gain Bandwidth: 3 MHz
- Low Quiescent Current: 475 µA per Amplifier
- High Common-Mode Rejection: 120 dB (typ)
- Low Input Bias Current: 8 pA
- Industry-Standard Package:
  - 5-Pin Small-Outline Transistor SOT-23 (DBV) Package

2 Applications

- Tracking Amplifier in Power Modules
- Merchant Power Supplies
- Transducer Amplifiers
- Bridge Amplifiers
- Temperature Measurements
- Strain Gauge Amplifiers
- Precision Integrators
- Battery-Powered Instruments
- Test Equipment

3 Description

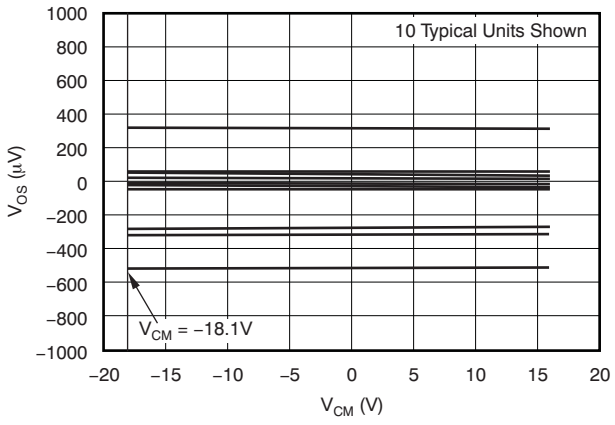
The OPA171-Q1 family of devices is a 36-V, single-supply, low-noise operational amplifier (op amp) with the ability to operate on supplies ranging from 2.7 V (±1.35 V) to 36 V (±18 V). This device is available in micro-packages and offers low offset, drift, and bandwidth with low quiescent current. The single, dual, and quad versions all have identical specifications for maximum design flexibility.

Device Information(1)

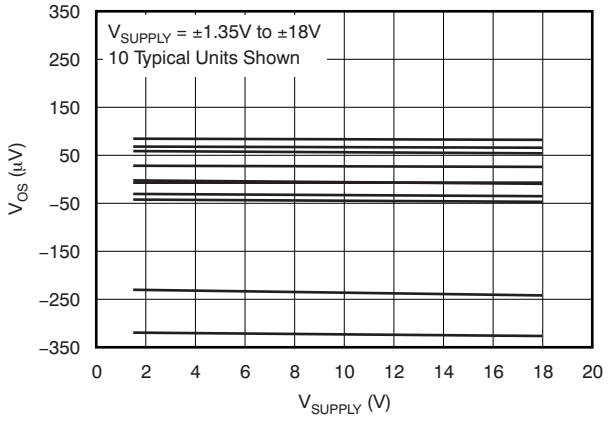
PART NUMBER	PACKAGE	BODY SIZE (NOM)
OPA171-Q1	SOT-23 (5)	2.90 mm × 1.60 mm
OPA2171-Q1	SOIC (8)	4.90 mm × 3.91 mm
	VSSOP (8)	3.00 mm × 3.00 mm
OPA4171-Q1	SOIC (14)	8.65 mm × 3.91 mm
	TSSOP (14)	5.00 mm × 4.40 mm

(1) For all available packages, see the orderable addendum at the end of the datasheet.

Offset Voltage vs Common-Mode Voltage



Offset Voltage vs Power Supply



An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

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4 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision B (December 2014) to Revision C — **Page**

- Changed the ESD classification levels for HBM and CDM in the *Features* list ... 1
- Added the 8-pin VSSOP (DGK) package option for the OPA2171-Q1 device ... 1
- Clarified the ESD values for each device in the *ESD Ratings* table ... 5

Changes from Revision A (September 2012) to Revision B — **Page**

- Added the *Handling Ratings* table, *Feature Description* section, *Device Functional Modes* section, *Application and Implementation* section, *Power Supply Recommendations* section, *Layout* section, *Device and Documentation Support* section, and *Mechanical, Packaging, and Orderable Information* section ... 1
- Added the OPA2171-Q1 and OPA4171-Q1 devices to the data sheet ... 1

Changes from Original (June, 2011) to Revision A — **Page**

- Added second bullet to Features: AEC-Q100 Test Guidance With the Following Results: –Device Temperature Grade1: -40°C to 125°C Ambient Operating Temperature Range –Device HBM ESD Classification Level H2 –Device CDM ESD Classification Level C3A ... 1
- Added classification levels to ESD ratings in Absolute Maximum Ratings table. ... 4
- Added row to Absolute Maximum Ratings table: Latch-up per JESD78D with Class 1 value. ... 4

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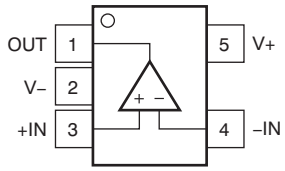
5 Description (continued)

Unlike most op amps, which are specified at only one supply voltage, the OPAx171-Q1 family of devices is specified from 2.7 to 36 V. Input signals beyond the supply rails do not cause phase reversal. The OPAx171-Q1 family of devices is stable with capacitive loads up to 300 pF. The input can operate 100 mV below the negative rail and within 2 V of the top rail during normal operation. The device can operate with full rail-to-rail input 100 mV beyond the top rail, but with reduced performance within 2 V of the top rail.

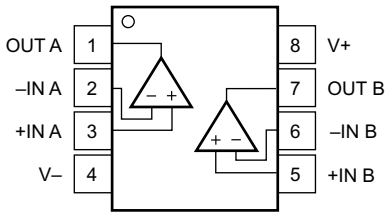
The OPAx171-Q1 op amp is specified from –40°C to +125°C.

6 Pin Configuration and Functions

DBV Package
5-Pin SOT23
OPA171-Q1 Top View



D or DGK Package
8-Pin SOIC or VSSOP
OPA2171-Q1 Top View

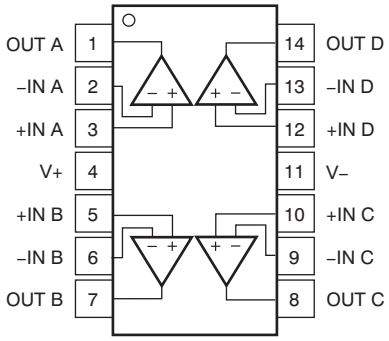


Pin Functions — SOT-23 (5), SOIC (8), and VSSOP (8) Packages

PIN			I/O	DESCRIPTION
NAME	OPA171-Q1 SOT-23 (5)	OPA2171-Q1 SOIC AND VSSOP (8)		
+IN	3	—	I	Noninverting input
+IN A	—	3	I	Noninverting input, channel A
+IN B	—	5	I	Noninverting input, channel B
–IN	4	—	I	Inverting input
–IN A	—	2	I	Inverting input, channel A
–IN B	—	6	I	Inverting input, channel B
OUT	1	—	O	Output
OUT A	—	1	O	Output, channel A
OUT B	—	7	O	Output, channel B
V+	5	7	—	Positive (highest) power supply
V–	2	4	—	Negative (lowest) power supply

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D and PW Packages
14-Pin SOIC and TSSOP
OPA4171-Q1 Top View



Pin Functions — SOIC (14) and TSSOP (14) Packages

PIN		I/O	DESCRIPTION
NAME	NO.		
+IN A	3	I	Noninverting input, channel A
+IN B	5	I	Noninverting input, channel B
+IN C	10	I	Noninverting input, channel C
+IN D	12	I	Noninverting input, channel D
–IN A	2	I	Inverting input, channel A
–IN B	6	I	Inverting input, channel B
–IN C	9	I	Inverting input, channel C
–IN D	13	I	Inverting input, channel D
OUT A	1	O	Output, channel A
OUT B	7	O	Output, channel B
OUT C	8	O	Output, channel C
OUT D	14	O	Output, channel D
V+	4	—	Positive (highest) power supply
V–	11	—	Negative (lowest) power supply

7 Specifications

7.1 Absolute Maximum Ratings

Over operating free-air temperature range, unless otherwise noted.(1)

		MIN	MAX	UNIT
Supply voltage			40	V
Signal input terminals	Voltage	(V–) – 0.5	(V+) + 0.5	V
	Current		±10	mA
Output short circuit(2)		Continuous		
Operating temperature		–55	150	°C
Junction temperature			150	°C
Latch-up per JESD78D		Class 1		
Storage temperature, T_{stg}		–65	150	°C

(1) Stresses above these ratings may cause permanent damage. Exposure to absolute maximum conditions for extended periods may degrade device reliability. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those specified is not implied.
(2) Short-circuit to ground, one amplifier per package.

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7.2 ESD Ratings

			VALUE	UNIT
OPA171-Q1				
$V_{(ESD)}$	Electrostatic discharge	Human body model (HBM), per AEC Q100-002 (1)	±4000	V
		Charged device model (CDM), per AEC Q100-011	±500	
OPA2171-Q1				
$V_{(ESD)}$	Electrostatic discharge	Human body model (HBM), per AEC Q100-002 (1)	±4000	V
		Charged device model (CDM), per AEC Q100-011	±1000	
OPA4171-Q1				
$V_{(ESD)}$	Electrostatic discharge	Human body model (HBM), per AEC Q100-002 (1)	±2000	V
		Charged device model (CDM), per AEC Q100-011	±1000	

(1) AEC Q100-002 indicates HBM stressing is done in accordance with the ANSI/ESDA/JEDEC JS-001 specification.

7.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

	MIN	NOM	MAX	UNIT
Supply voltage (V+ – V–)	4.5 (±2.25)		36 (±18)	V
Specified temperature	–40		125	°C

7.4 Thermal Information — OPA171-Q1 and OPA2171-Q1

THERMAL METRIC (1)		OPA171-Q1	OPA2171-Q1		UNIT
		DBV (SOT-23)	D (SOIC)	DGK (VSSOP)	
		5 PINS	8 PINS	8 PINS	
$R_{\theta JA}$	Junction-to-ambient thermal resistance	277.3	116.1	186.5	°C/W
$R_{\theta JC(top)}$	Junction-to-case(top) thermal resistance	193.3	69.8	78	°C/W
$R_{\theta JB}$	Junction-to-board thermal resistance	121.2	56.6	107.8	°C/W
ψ_{JT}	Junction-to-top characterization parameter	51.8	22.5	15.6	°C/W
ψ_{JB}	Junction-to-board characterization parameter	109.5	56.1	106.2	°C/W

(1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report, SPRA953.

7.5 Thermal Information — OPA4171-Q1

THERMAL METRIC (1)		OPA4171-Q1		UNIT
		D (SOIC)	PW (TSSOP)	
		14 PINS	14 PINS	
$R_{\theta JA}$	Junction-to-ambient thermal resistance	93.2	106.9	°C/W
$R_{\theta JC(top)}$	Junction-to-case(top) thermal resistance	51.8	24.4	°C/W
$R_{\theta JB}$	Junction-to-board thermal resistance	49.4	59.3	°C/W
ψ_{JT}	Junction-to-top characterization parameter	13.5	0.6	°C/W
ψ_{JB}	Junction-to-board characterization parameter	42.2	54.3	°C/W

(1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report, SPRA953.

7.6 Electrical Characteristics

At T_A = 25°C, V_S = 2.7 to 36 V, V_{CM} = V_{OUT} = V_S / 2, and R_{LOAD} = 10 kΩ connected to V_S / 2, unless otherwise noted. The specified temperature range is T_A = –40°C to +125°C.

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
OFFSET VOLTAGE						
V_{OS}	Input offset voltage			0.25	±1.8	mV
	Input offset voltage over temperature			0.3	±2	mV

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Electrical Characteristics (continued)

At T_A = 25°C, V_S = 2.7 to 36 V, V_{CM} = V_{OUT} = V_S / 2, and R_{LOAD} = 10 kΩ connected to V_S / 2, unless otherwise noted. The specified temperature range is T_A = –40°C to +125°C.

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
dV_{OS}/dT	Input offset voltage drift (over temperature)			0.3	±2(1)	µV/°C
PSRR	Input offset voltage over temperature vs power supply	V_S = 4 to 36 V		1	±3	µV/V
	Channel separation, DC			5		µV/V

(1) Not production tested.

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Electrical Characteristics (continued)

At T_A = 25°C, V_S = 2.7 to 36 V, V_{CM} = V_{OUT} = V_S / 2, and R_{LOAD} = 10 kΩ connected to V_S / 2, unless otherwise noted. The specified temperature range is T_A = –40°C to +125°C.

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
INPUT BIAS CURRENT						
I_B	Input bias current			±8	±15	pA
	Input bias current over temperature				±3.5	nA
I_{OS}	Input offset current			±4		pA
	Input offset current over temperature				±3.5	nA
NOISE						
	Input voltage noise	f = 0.1 Hz to 10 Hz		3		μV_{PP}
e_n	Input voltage noise density	f = 100 Hz		25		nV/√Hz
		f = 1 kHz		14		nV/√Hz
INPUT VOLTAGE						
V_{CM}	Common-mode voltage range[(2)]		(V–) – 0.1		(V+) – 2	V
CMRR	Common-mode rejection ratio (over temperature)	V_S = ±2 V, (V–) – 0.1 V < V_{CM} < (V+) – 2 V	90	104		dB
		V_S = ±18 V, (V–) – 0.1 V < V_{CM} < (V+) – 2 V	104	120		dB
INPUT IMPEDANCE						
	Differential			100 || 3		MΩ || pF
	Common-mode			6 || 3		10^{12}Ω || pF
OPEN-LOOP GAIN						
A_{OL}	Open-loop voltage gain (over temperature)	V_S = +4V to +36V, (V–) + 0.35V < V_O < (V+) – 0.35V	110	130		dB
FREQUENCY RESPONSE						
GBP	Gain bandwidth product			3.0		MHz
SR	Slew rate	G = 1		1.5		V/µs
t_S	Settling time	To 0.1%, V_S = ±18 V, G = 1, 10-V step		6		µs
		To 0.01% (12 bit), V_S = ±18 V, G = 1, 10-V step		10		µs
	Overload recovery time	$V_{±IN}$ × Gain > V_S		2		µs
THD+N	Total harmonic distortion + noise	G = 1, f = 1 kHz, V_O = 3 V_{RMS}		0.0002%		
OUTPUT						
V_O	Voltage output swing from rail (over temperature)	R_L = 10 kΩ, A_{OL} ≥ 110 dB	(V–) + 0.35		(V+) – 0.35	V
I_{SC}	Short-circuit current	Sourcing		25		mA
		Sinking		–35		
C_{LOAD}	Capacitive load drive		See the *Typical Characteristics* section			pF
R_O	Open-loop output resistance	f = 1 MHz, I_O = 0 A		150		Ω
POWER SUPPLY						
V_S	Specified voltage range		2.7		36	V
I_Q	Quiescent current per amplifier	I_O = 0 A		475	595	µA
	Quiescent current per amplifier (over temperature)	I_O = 0 A			650	µA
TEMPERATURE						
	Specified range		–40		125	°C
	Operating range		–55		150	°C

(2) The input range can be extended beyond (V+) – 2 V up to V+. See the *Typical Characteristics* and *Detailed Description* sections for additional information.

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7.7 Typical Characteristics

V_S = ±18 V, V_{CM} = V_S / 2, R_{LOAD} = 10 kΩ connected to V_S / 2, and C_L = 100 pF, unless otherwise noted.

Table 1. Characteristic Performance Measurements

DESCRIPTION	FIGURE
Offset Voltage Production Distribution	Figure 1
Offset Voltage Drift Distribution	Figure 2
Offset Voltage vs Temperature	Figure 3
Offset Voltage vs Common-Mode Voltage	Figure 4
Offset Voltage vs Common-Mode Voltage (Upper Stage)	Figure 5
Offset Voltage vs Power Supply	Figure 6
I_B and I_{OS} vs Common-Mode Voltage	Figure 7
Input Bias Current vs Temperature	Figure 8
Output Voltage Swing vs Output Current (Maximum Supply)	Figure 9
CMRR and PSRR vs Frequency (Referred-to Input)	Figure 10
CMRR vs Temperature	Figure 11
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0.1Hz to 10Hz Noise	Figure 13
Input Voltage Noise Spectral Density vs Frequency	Figure 14
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THD+N vs Output Amplitude	Figure 16
Quiescent Current vs Temperature	Figure 17
Quiescent Current vs Supply Voltage	Figure 18
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Large-Signal Settling Time (10-V Positive Step)	Figure 32
Large-Signal Settling Time (10-V Negative Step)	Figure 33
Short-Circuit Current vs Temperature	Figure 34
Maximum Output Voltage vs Frequency	Figure 35
Channel Separation vs Frequency	Figure 36

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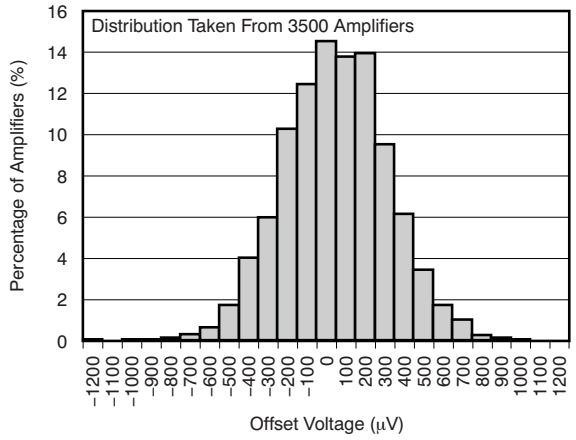


Figure 1. Offset Voltage Production Distribution

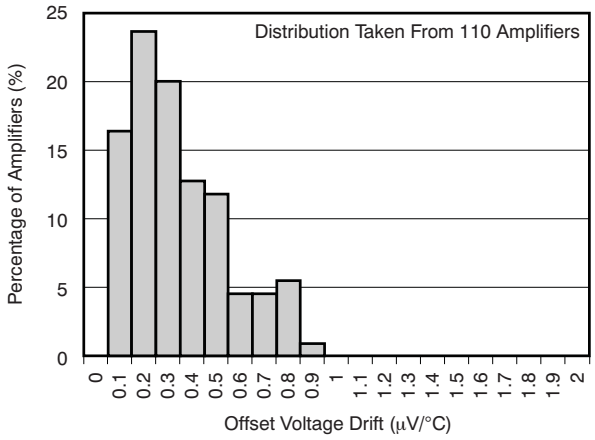


Figure 2. Offset Voltage Drift Distribution

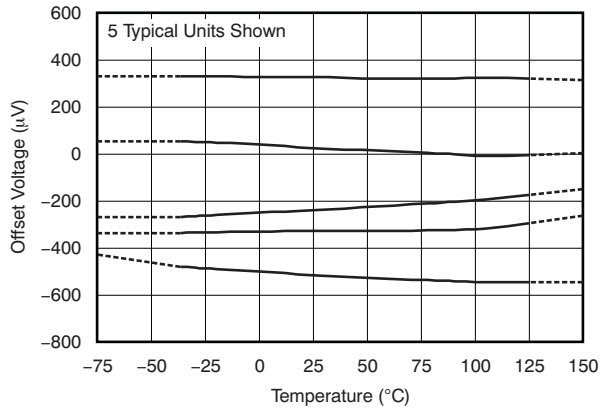


Figure 3. Offset Voltage vs Temperature

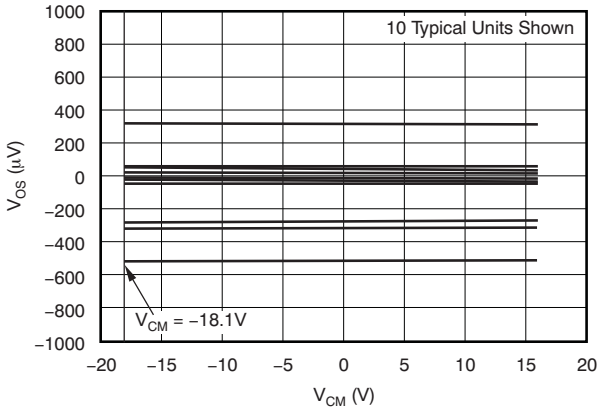


Figure 4. Offset Voltage vs Common-Mode Voltage

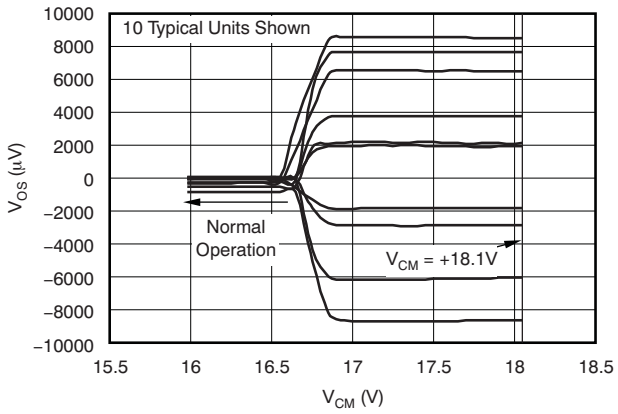


Figure 5. Offset Voltage vs Common-Mode Voltage (Upper Stage)

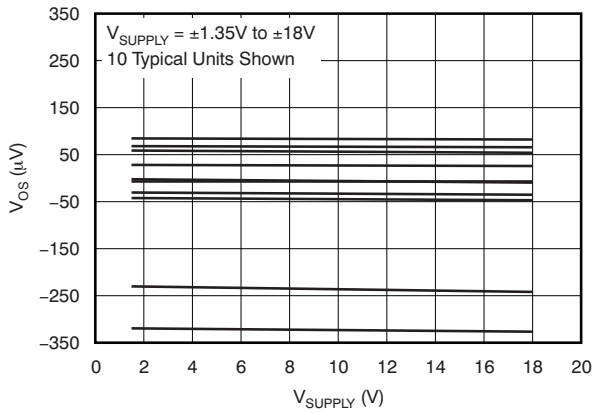


Figure 6. Offset Voltage vs Power Supply

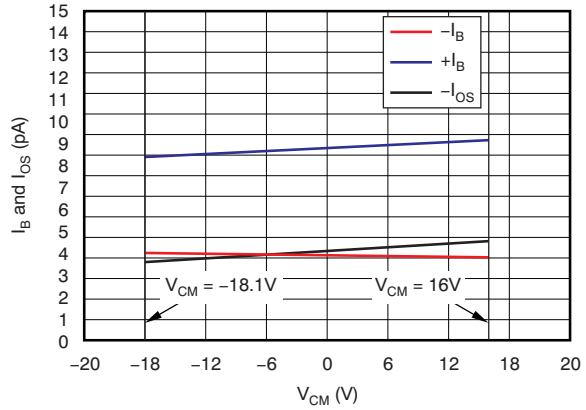


Figure 7. I_B and I_{OS} vs Common-Mode Voltage

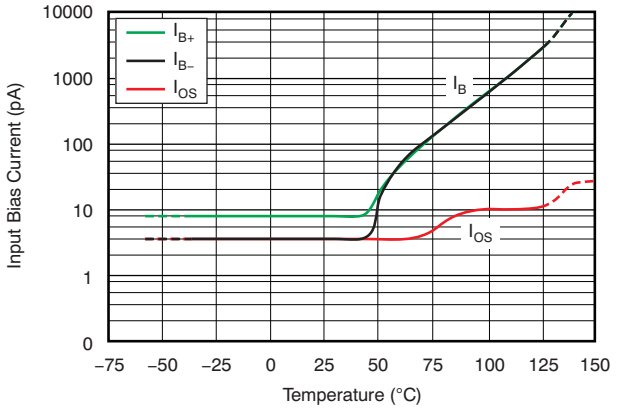


Figure 8. Input Bias Current vs Temperature

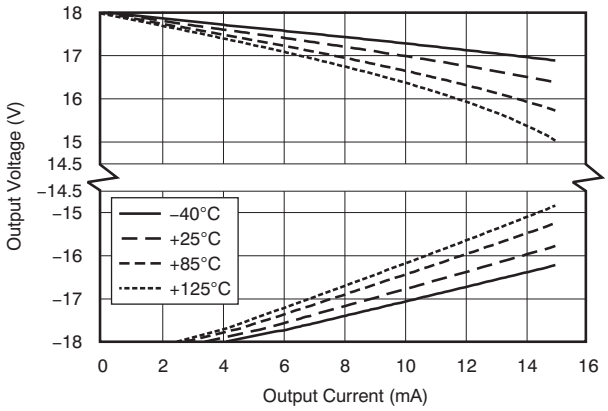


Figure 9. Output Voltage Swing vs Output Current (Maximum Supply)

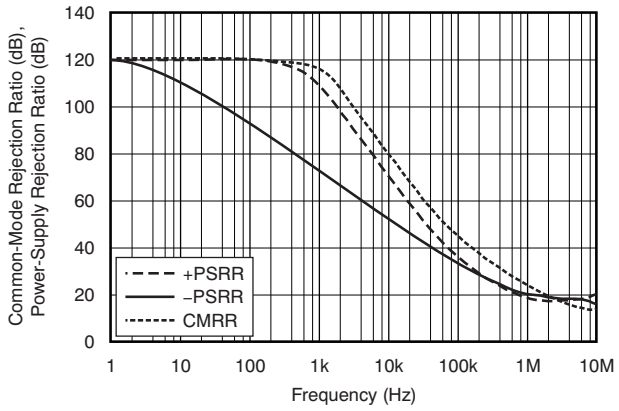


Figure 10. CMRR and PSRR vs Frequency (Referred-to Input)

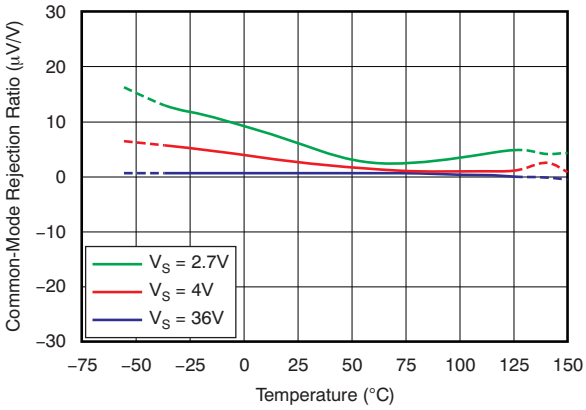


Figure 11. CMRR vs Temperature

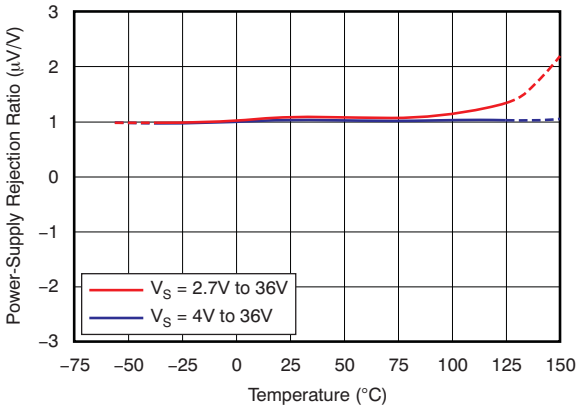


Figure 12. PSRR vs Temperature

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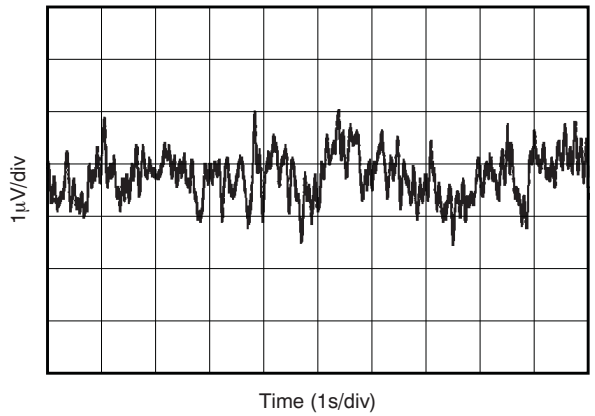


Figure 13. 0.1- to 10-Hz Noise

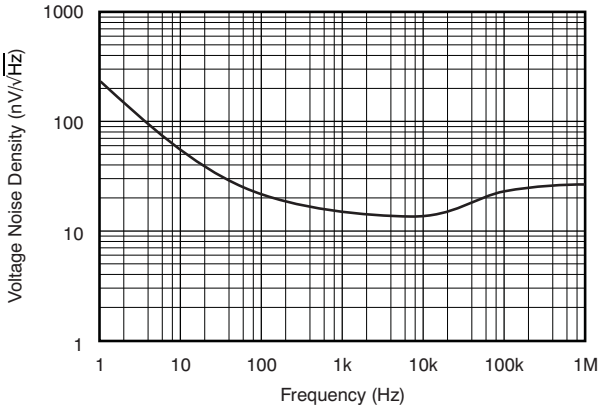


Figure 14. Input Voltage Noise Spectral Density vs Frequency

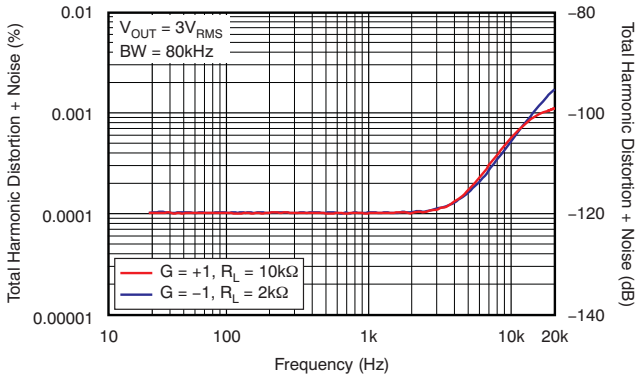


Figure 15. THD+N Ratio vs Frequency

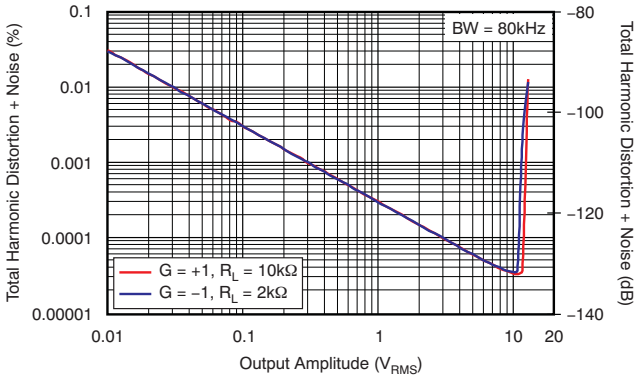


Figure 16. THD+N vs Output Amplitude

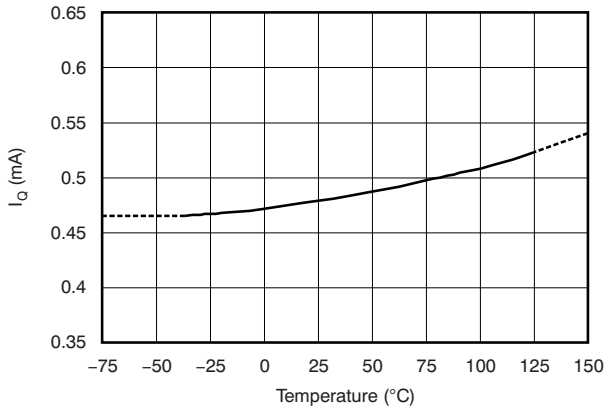


Figure 17. Quiescent Current vs Temperature

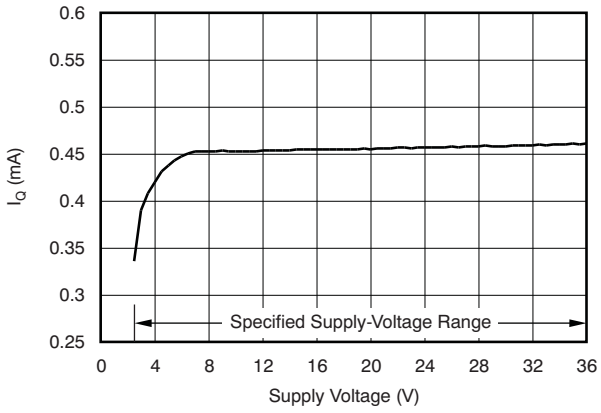


Figure 18. Quiescent Current vs Supply Voltage

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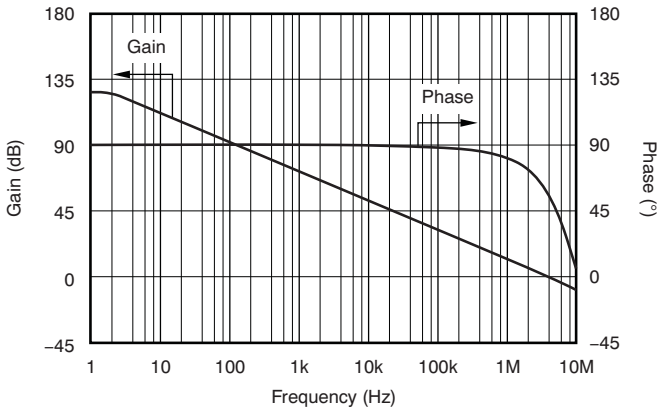


Figure 19. Open-Loop Gain and Phase vs Frequency

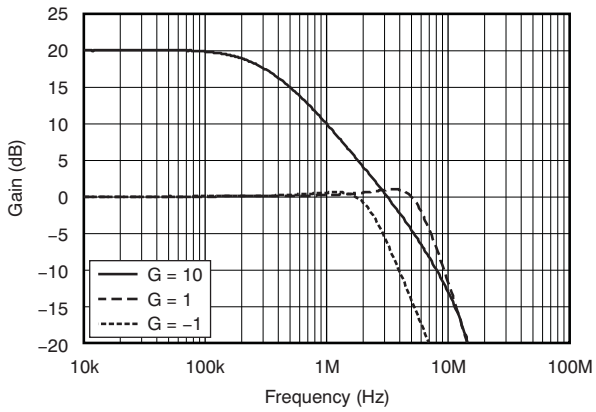


Figure 20. Closed-Loop Gain vs Frequency

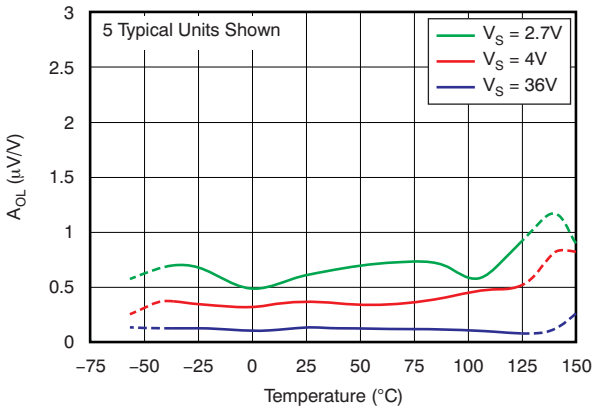


Figure 21. Open-Loop Gain vs Temperature

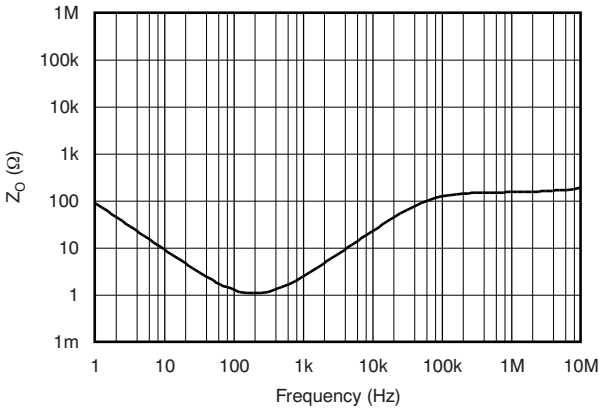


Figure 22. Open-Loop Output Impedance vs Frequency

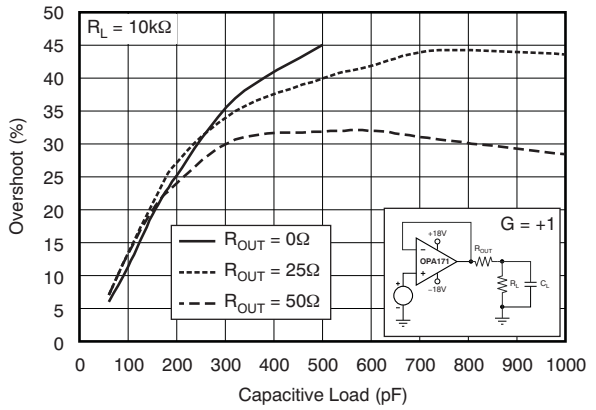


Figure 23. Small-Signal Overshoot vs Capacitive Load (100-mV Output Step)

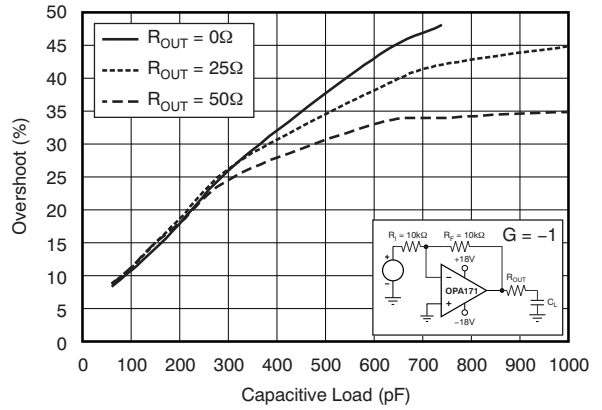


Figure 24. Small-Signal Overshoot vs Capacitive Load (100-mV Output Step)

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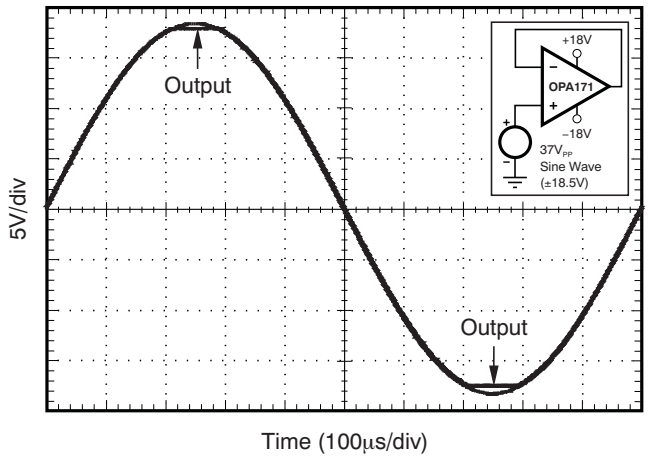


Figure 25. No Phase Reversal

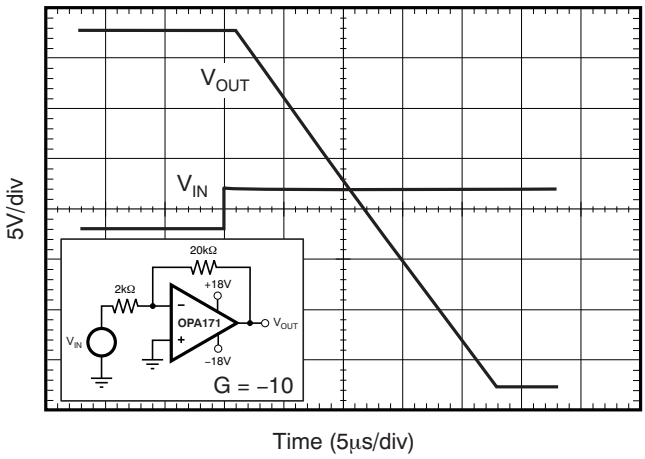


Figure 26. Positive Overload Recovery

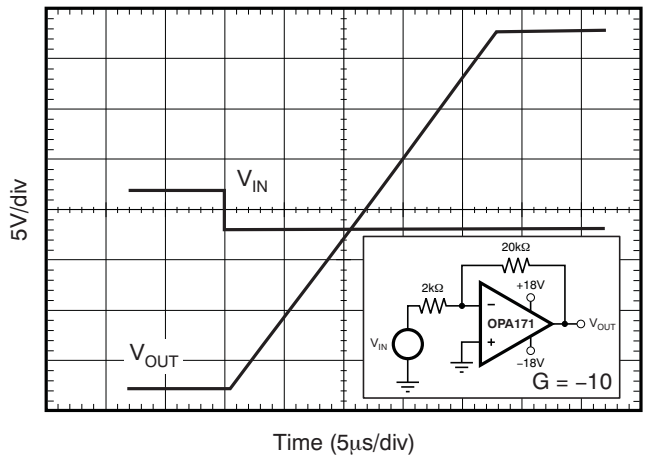


Figure 27. Negative Overload Recovery

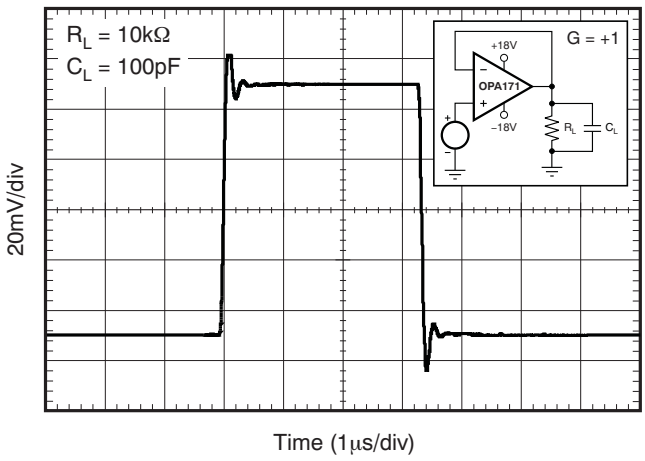


Figure 28. Small-Signal Step Response (100 mV)

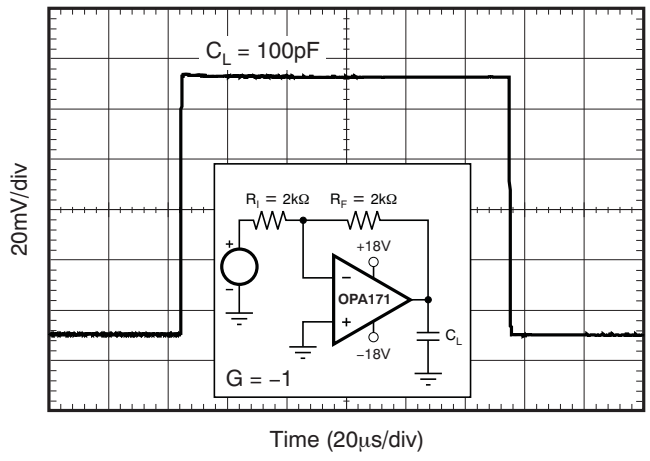


Figure 29. Small-Signal Step Response (100 mV)

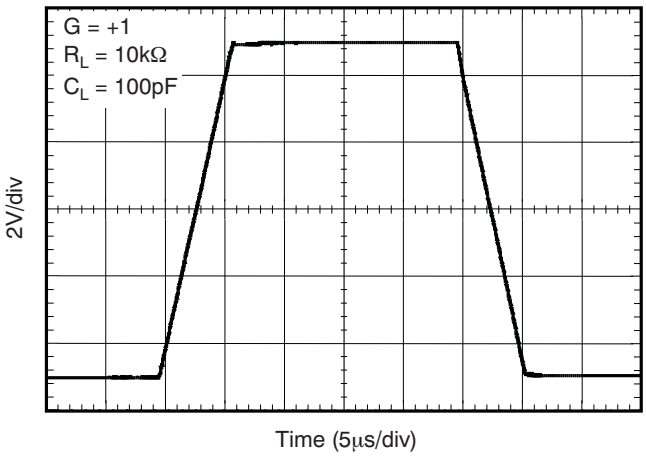


Figure 30. Large-Signal Step Response

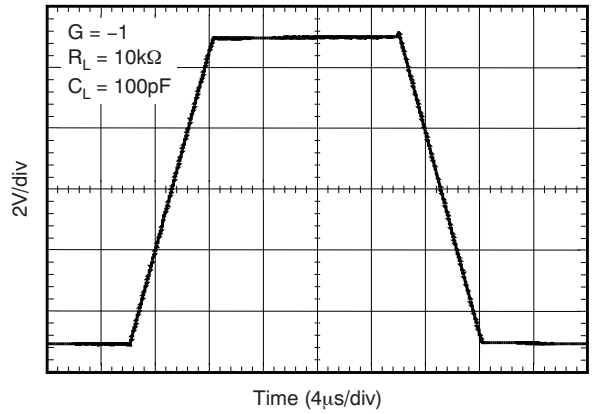


Figure 31. Large-Signal Step Response

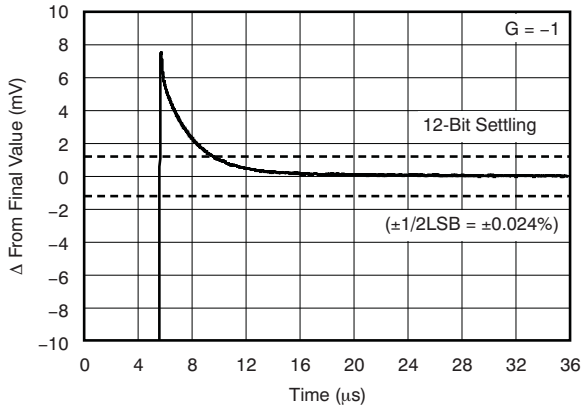


Figure 32. Large-Signal Settling Time (10-V Positive Step)

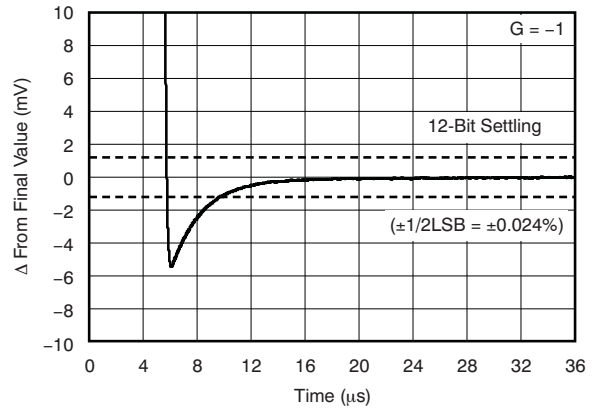


Figure 33. Large-Signal Settling Time (10-V Negative Step)

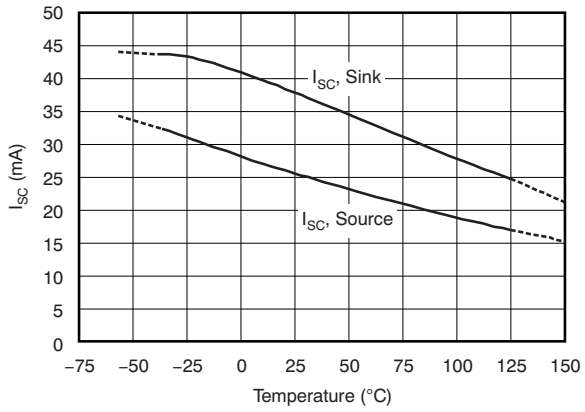


Figure 34. Short-Circuit Current vs Temperature

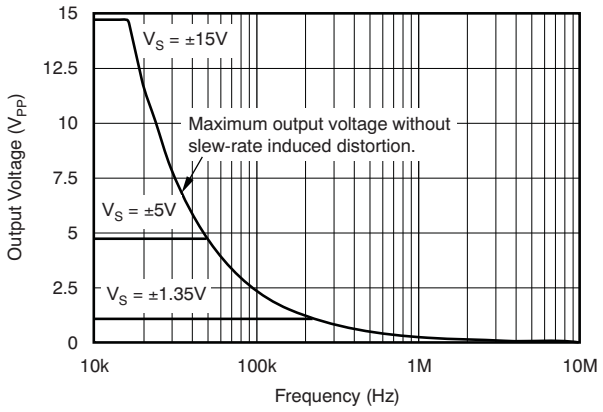


Figure 35. Maximum Output Voltage vs Frequency

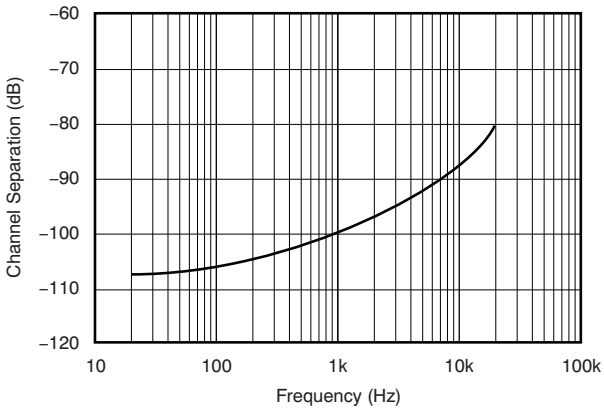


Figure 36. Channel Separation vs Frequency

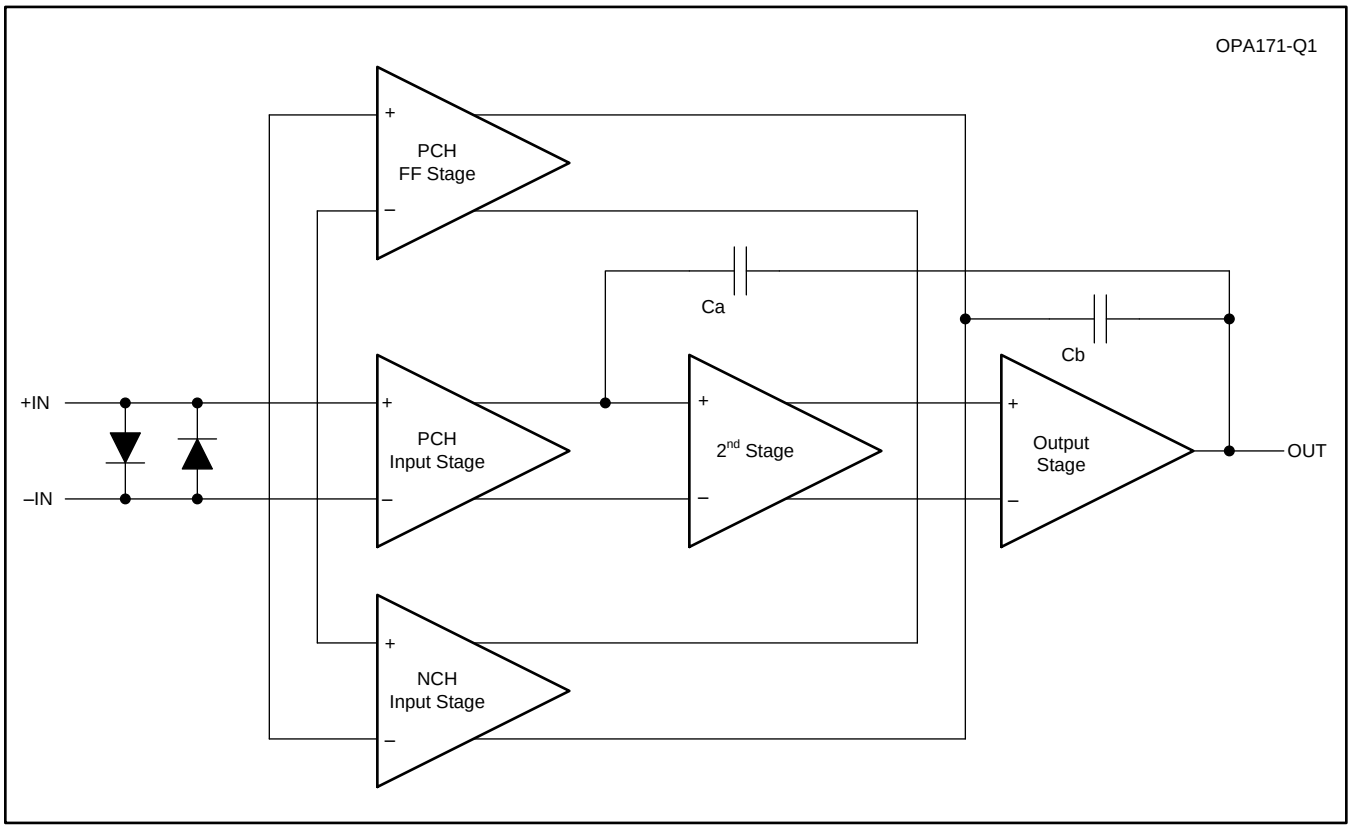
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8 Detailed Description

8.1 Overview

The OPAx171-Q1 family of operational amplifiers provide high overall performance, making them ideal for many general-purpose applications. The excellent offset drift of only 1.5 µV/°C (maximum) provides excellent stability over the entire temperature range. In addition, the device offers very good overall performance with high CMRR, PSRR, AOL, and superior THD.

8.2 Functional Block Diagram



8.3 Feature Description

8.3.1 Operating Characteristics

The OPAx171-Q1 family of devices is specified for operation from 2.7 to 36 V (±1.35 to ±18 V). Many of the specifications apply from –40°C to +125°C. Parameters that can exhibit significant variance with regard to operating voltage or temperature are shown in the *Typical Characteristics* section.

8.3.2 Phase-Reversal Protection

The OPAx171-Q1 family of devices has an internal phase-reversal protection. Many op amps exhibit a phase reversal when the input is driven beyond its linear common-mode range. This condition is most often encountered in noninverting circuits when the input is driven beyond the specified common-mode voltage range, causing the output to reverse into the opposite rail. The input of the OPAx171-Q1 family of devices prevents phase reversal with excessive common-mode voltage. Instead, the output limits into the appropriate rail. Figure 37 shows this performance.

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Feature Description (continued)

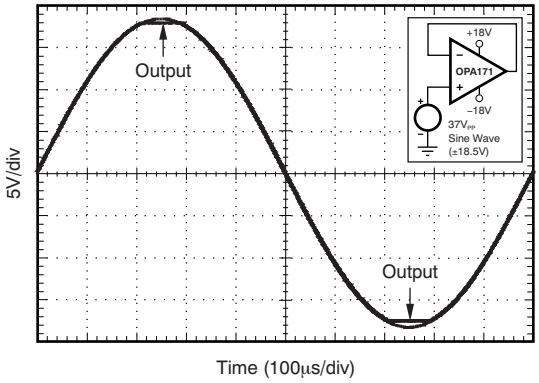


Figure 37. No Phase Reversal

8.3.3 Capacitive Load and Stability

The dynamic characteristics of the OPAx171-Q1 family of devices have been optimized for commonly encountered operating conditions. The combination of low closed-loop gain and high capacitive loads decreases the phase margin of the amplifier and can lead to gain peaking or oscillations. As a result, heavier capacitive loads must be isolated from the output. The simplest way to achieve this isolation is to add a small resistor (for example, R_{OUT} equal to 50 Ω) in series with the output. Figure 38 and Figure 39 show small-signal overshoot versus capacitive load for several values of R_{OUT}. Also, for details of analysis techniques and application circuits, refer to the *Applications Bulletin AB-028* (SBOA015), available for download from TI.com.

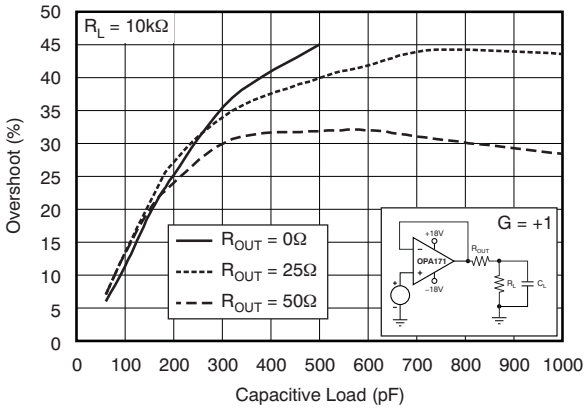


Figure 38. Small-Signal Overshoot versus Capacitive Load (100-mV Output Step)

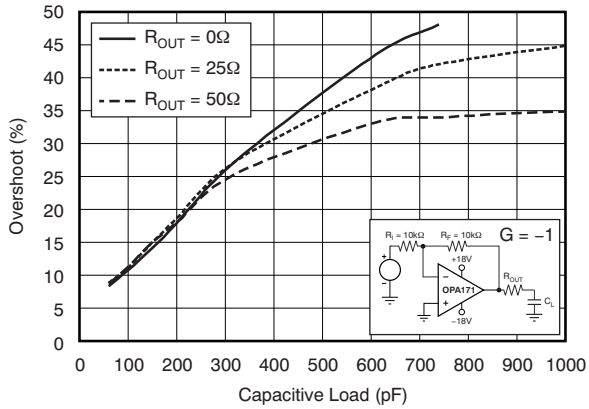


Figure 39. Small-Signal Overshoot versus Capacitive Load (100-mV Output Step)

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8.4 Device Functional Modes

8.4.1 Common-Mode Voltage Range

The input common-mode voltage range of the OPAx171-Q1 family of devices extends 100 mV below the negative rail and within 2 V of the top rail for normal operation.

This device can operate with full rail-to-rail input 100 mV beyond the top rail, but with reduced performance within 2 V of the top rail. The typical performance in this range is summarized in Table 2.

Table 2. Typical Performance Range

PARAMETER	MIN	TYP	MAX	UNIT
Input common-mode voltage	(V+) – 2		(V+) + 0.1	V
Offset voltage		7		mV
Offset voltage vs temperature		12		µV/°C
Common-mode rejection		65		dB
Open-loop gain		60		dB
GBW		0.7		MHz
Slew rate		0.7		V/µs
Noise at f = 1kHz		30		nV/$\sqrt{\text{Hz}}$

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9 Application and Implementation

NOTE

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

9.1 Application Information

The OPAx171-Q1 operational amplifier provides high overall performance, making it ideal for many general-purpose applications. The excellent offset drift of only 2 µV/°C provides excellent stability over the entire temperature range. In addition, the device offers very-good overall performance with high CMRR, PSRR, and A_{OL}. As with all amplifiers, applications with noisy or high-impedance power supplies require decoupling capacitors close to the device pins. In most cases, 0.1-µF capacitors are adequate.

9.1.1 Electrical Overstress

Designers often ask questions about the capability of an op amp to withstand electrical overstress. These questions tend to focus on the device inputs, but may involve the supply voltage pins or even the output pin. Each of these different pin functions have electrical stress limits determined by the voltage breakdown characteristics of the particular semiconductor fabrication process and specific circuits connected to the pin. Additionally, internal electrostatic discharge (ESD) protection is built into these circuits to protect them from accidental ESD events both before and during product assembly.

These ESD protection diodes also provide in-circuit, input overdrive protection, as long as the current is limited to 10 mA as stated in the *Absolute Maximum Ratings* table. Figure 40 shows how a series input resistor can be added to the driven input to limit the input current. The added resistor contributes thermal noise at the amplifier input and its value should be kept to a minimum in noise-sensitive applications.

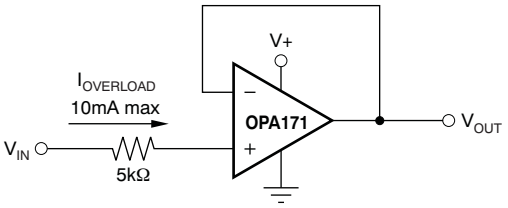


Figure 40. Input Current Protection

An ESD event produces a short duration, high-voltage pulse that is transformed into a short duration, high-current pulse as it discharges through a semiconductor device. The ESD protection circuits are designed to provide a current path around the operational amplifier core to prevent it from being damaged. The energy absorbed by the protection circuitry is then dissipated as heat.

When the operational amplifier connects into a circuit, the ESD protection components are intended to remain inactive and not become involved in the application circuit operation. However, circumstances may arise where an applied voltage exceeds the operating voltage range of a given pin. Should this condition occur, there is a risk that some of the internal ESD protection circuits may be biased on, and conduct current. Any such current flow occurs through ESD cells and rarely involves the absorption device.

If the ability of the supply to absorb this current is uncertain, external zener diodes may be added to the supply pins. The zener voltage must be selected such that the diode does not turn on during normal operation.

However, the zener voltage should be low enough so that the zener diode conducts if the supply pin begins to rise above the safe operating supply voltage level.

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9.2 Typical Application

9.2.1 Capacitive Load Drive Solution Using an Isolation Resistor

The OPA171-Q1 device can be used capacitive loads such as cable shields, reference buffers, MOSFET gates, and diodes. The circuit uses an isolation resistor (R_{ISO}) to stabilize the output of an op amp. R_{ISO} modifies the open loop gain of the system to ensure the circuit has sufficient phase margin.

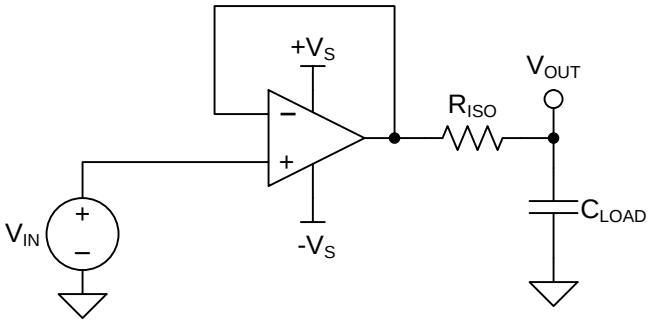


Figure 41. Unity-Gain Buffer with R_{ISO} Stability Compensation

9.2.1.1 Design Requirements

The design requirements are:

- Supply voltage: 30 V (±15 V)
- Capacitive loads: 100 pF, 1000 pF, 0.01 µF, 0.1 µF, and 1 µF
- Phase margin: 45° and 60°

9.2.1.2 Detailed Design Procedure

Figure 42 shows a unity-gain buffer driving a capacitive load. Equation 1 shows the transfer function for the circuit in Figure 42. Not shown in Figure 42 is the open-loop output resistance of the op amp, R_o.

$$T(s) = \frac{1 + C_{LOAD} \times R_{ISO} \times s}{1 + \left(R_o + R_{ISO}\right) \times C_{LOAD} \times s} \quad (1)$$

The transfer function in Equation 1 has a pole and a zero. The frequency of the pole (f_p) is determined by (R_o + R_{ISO}) and C_{LOAD}. Components R_{ISO} and C_{LOAD} determine the frequency of the zero (f_z). A stable system is obtained by selecting R_{ISO} such that the rate of closure (ROC) between the open-loop gain (A_{OL}) and 1/β is 20 dB/decade. Figure 42 shows the concept. The 1/β curve for a unity-gain buffer is 0 dB.

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Typical Application (continued)

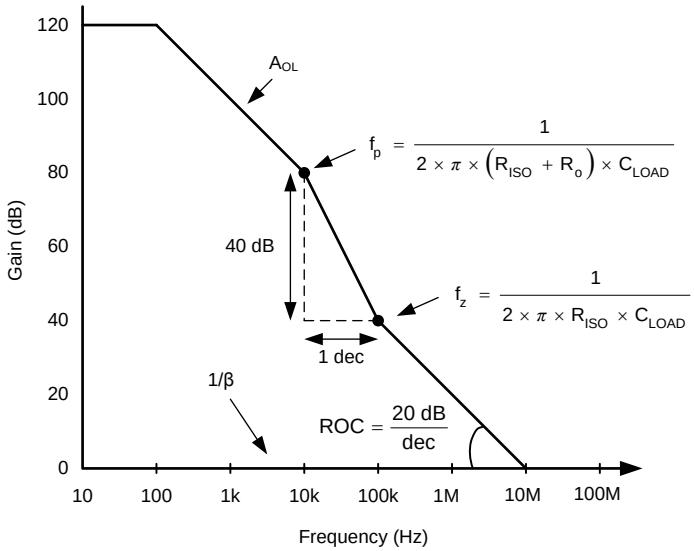


Figure 42. Unity-Gain Amplifier with R_{ISO} Compensation

ROC stability analysis is typically simulated. The validity of the analysis depends on multiple factors, especially the accurate modeling of R_o. In addition to simulating the ROC, a robust stability analysis includes a measurement of overshoot percentage and AC gain peaking of the circuit using a function generator, oscilloscope, and gain and phase analyzer. Phase margin is then calculated from these measurements. Table 3 lists the overshoot percentage and AC gain peaking that correspond to phase margins of 45° and 60°. For more details on this design and other alternative devices that can be used in place of the OPA171, refer to the Precision Design, *Capacitive Load Drive Solution using an Isolation Resistor* (TIPD128).

Table 3. Phase Margin versus Overshoot and AC Gain Peaking

PHASE MARGIN	OVERSHOOT	AC GAIN PEAKING
45°	23.3%	2.35 dB
60°	8.8%	0.28 dB

9.2.1.3 Application Curve

The OPA171-Q1 device meets the supply voltage requirements of 30 V. The OPA171-Q1 device was tested for various capacitive loads and R_{ISO} was adjusted to achieve an overshoot corresponding to Table 3. Figure 43 shows the test results.

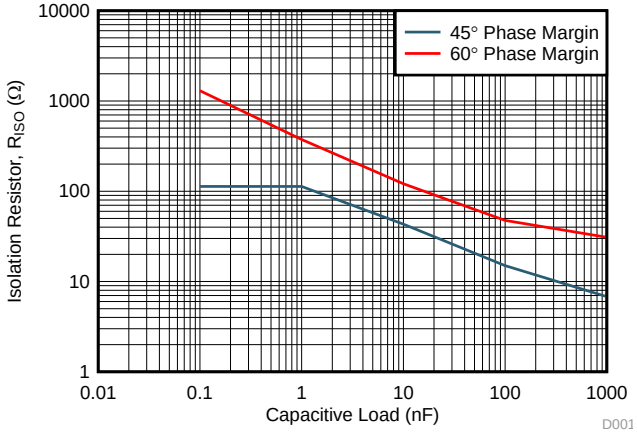


Figure 43. R_{ISO} vs C_{LOAD}

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10 Power Supply Recommendations

The OPAx171-Q1 family of devices is specified for operation from 4.5 V to 36 V (±2.25 V to ±18 V); many specifications apply from –40°C to +125°C. Parameters that can exhibit significant variance with regard to operating voltage or temperature are presented in the *Typical Characteristics* section.

CAUTION

Supply voltages larger than 40 V can permanently damage the device; see the *Absolute Maximum Ratings* table.

Place 0.1-µF bypass capacitors close to the power-supply pins to reduce errors coupling in from noisy or high-impedance power supplies. For detailed information on bypass capacitor placement, see the *Layout* section.

11 Layout

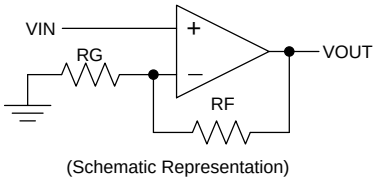
11.1 Layout Guidelines

For best operational performance of the device, use good printed circuit board (PCB) layout practices, including:

- Noise can propagate into analog circuitry through the power pins of the circuit as a whole and op amp itself. Bypass capacitors are used to reduce the coupled noise by providing low-impedance power sources local to the analog circuitry.
  - Connect low-ESR, 0.1-µF ceramic bypass capacitors between each supply pin and ground, placed as close to the device as possible. A single bypass capacitor from V+ to ground is applicable for single-supply applications.
- Separate grounding for analog and digital portions of circuitry is one of the simplest and most-effective methods of noise suppression. One or more layers on multilayer PCBs are usually devoted to ground planes. A ground plane helps distribute heat and reduces EMI noise pickup. Make sure to physically separate digital and analog grounds paying attention to the flow of the ground current. See *Circuit Board Layout Techniques*, SLOA089, for detailed information.
- In order to reduce parasitic coupling, run the input traces as far away from the supply or output traces as possible. If it is not possible to keep them separate, it is much better to cross the sensitive trace perpendicular as opposed to in parallel with the noisy trace.
- Place the external components as close to the device as possible. As shown in Figure 44, keeping RF and RG close to the inverting input minimizes parasitic capacitance.
- Keep the length of input traces as short as possible. Always remember that the input traces are the most sensitive part of the circuit.
- Consider a driven, low-impedance guard ring around the critical traces. A guard ring can significantly reduce leakage currents from nearby traces that are at different potentials.

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11.2 Layout Example



(Schematic Representation)

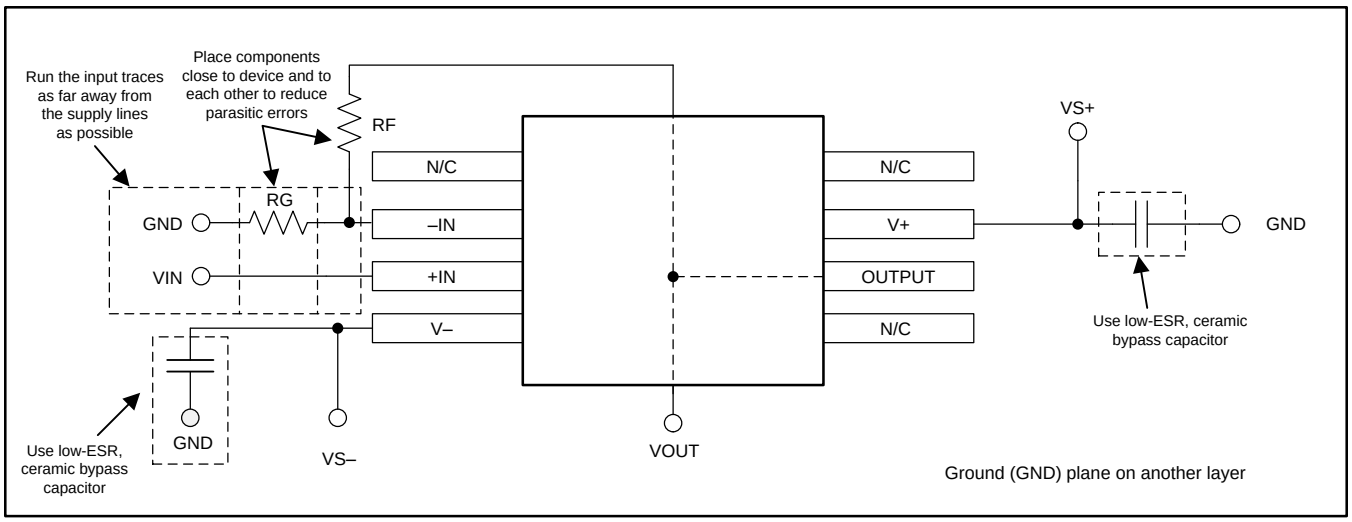


Figure 44. Operational Amplifier Board Layout for Noninverting Configuration

12 Device and Documentation Support

12.1 Documentation Support

12.1.1 Related Documentation

For related documentation see the following:

- *Applications Bulletin AB-028*, SBOA015
- *Capacitive Load Drive Solution using an Isolation Resistor*, TIDU032
- *Circuit Board Layout Techniques*, SLOA089

12.2 Related Links

The table below lists quick access links. Categories include technical documents, support and community resources, tools and software, and quick access to sample or buy.

Table 4. Related Links

PARTS	PRODUCT FOLDER	SAMPLE & BUY	TECHNICAL DOCUMENTS	TOOLS & SOFTWARE	SUPPORT & COMMUNITY
OPA171-Q1	Click here	Click here	Click here	Click here	Click here
OPA2171-Q1	Click here	Click here	Click here	Click here	Click here
OPA4171-Q1	Click here	Click here	Click here	Click here	Click here

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12.3 Community Resource

The following links connect to TI community resources. Linked contents are provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's Terms of Use.

TI E2E™ Online Community ***TI's Engineer-to-Engineer (E2E) Community.*** Created to foster collaboration among engineers. At e2e.ti.com, you can ask questions, share knowledge, explore ideas and help solve problems with fellow engineers.

Design Support ***TI's Design Support*** Quickly find helpful E2E forums along with design support tools and contact information for technical support.

12.4 Trademarks

E2E is a trademark of Texas Instruments.
All other trademarks are the property of their respective owners.

12.5 Electrostatic Discharge Caution



These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

12.6 Glossary

SLYZ022 — *TI Glossary*.

This glossary lists and explains terms, acronyms, and definitions.

13 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

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PACKAGING INFORMATION

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead/Ball Finish (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
OPA171AQDBVRQ1	ACTIVE	SOT-23	DBV	5	3000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 125	OULQ	Samples
OPA2171AQDGKRQ1	ACTIVE	VSSOP	DGK	8	2500	Green (RoHS & no Sb/Br)	CU NIPDAUAG	Level-2-260C-1 YEAR	-40 to 125	2171	Samples
OPA2171AQDRQ1	ACTIVE	SOIC	D	8	2500	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-3-260C-168 HR	-40 to 125	2171AQ	Samples
OPA4171AQDRQ1	ACTIVE	SOIC	D	14	2500	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-3-260C-168 HR	-40 to 125	OPA4171Q1	Samples
OPA4171AQPWRQ1	ACTIVE	TSSOP	PW	14	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-3-260C-168 HR	-40 to 125	O4171Q1	Samples

(1) The marketing status values are defined as follows:
ACTIVE: Product device recommended for new designs.
LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.
NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.
PREVIEW: Device has been announced but is not in production. Samples may or may not be available.
OBSOLETE: TI has discontinued the production of the device.

(2) Eco Plan - The planned eco-friendly classification: Pb-Free (RoHS), Pb-Free (RoHS Exempt), or Green (RoHS & no Sb/Br) - please check http://www.ti.com/productcontent for the latest availability information and additional product content details.
TBD: The Pb-Free/Green conversion plan has not been defined.
Pb-Free (RoHS): TI's terms "Lead-Free" or "Pb-Free" mean semiconductor products that are compatible with the current RoHS requirements for all 6 substances, including the requirement that lead not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, TI Pb-Free products are suitable for use in specified lead-free processes.
Pb-Free (RoHS Exempt): This component has a RoHS exemption for either 1) lead-based flip-chip solder bumps used between the die and package, or 2) lead-based die adhesive used between the die and leadframe. The component is otherwise considered Pb-Free (RoHS compatible) as defined above.
Green (RoHS & no Sb/Br): TI defines "Green" to mean Pb-Free (RoHS compatible), and free of Bromine (Br) and Antimony (Sb) based flame retardants (Br or Sb do not exceed 0.1% by weight in homogeneous material)

(3) MSL, Peak Temp. - The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

(4) There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.

(5) Multiple Device Markings will be inside parentheses. Only one Device Marking contained in parentheses and separated by a "~" will appear on a device. If a line is indented then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.

(6) Lead/Ball Finish - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead/Ball Finish values may wrap to two lines if the finish value exceeds the maximum column width.

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In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

OTHER QUALIFIED VERSIONS OF OPA171-Q1, OPA2171-Q1, OPA4171-Q1 :

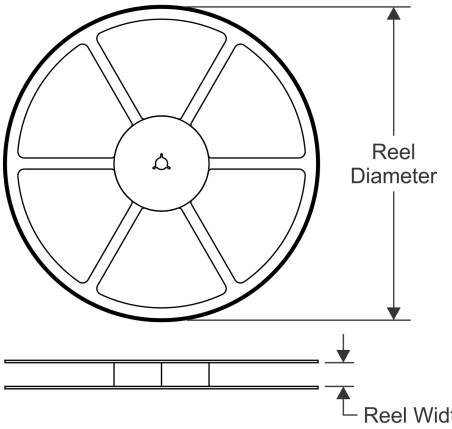
- Catalog: OPA171, OPA2171, OPA4171
- Enhanced Product: OPA2171-EP

NOTE: Qualified Version Definitions:

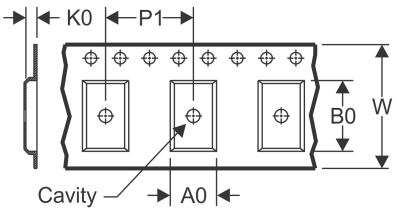
- Catalog - TI's standard catalog product
- Enhanced Product - Supports Defense, Aerospace and Medical Applications

TAPE AND REEL INFORMATION

REEL DIMENSIONS

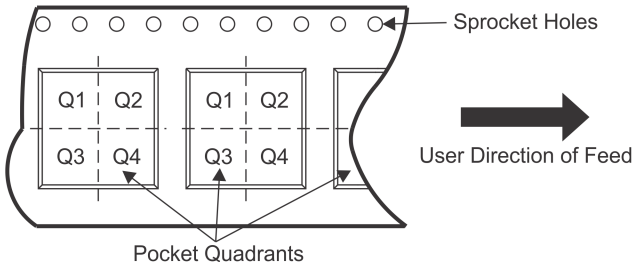


TAPE DIMENSIONS



A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

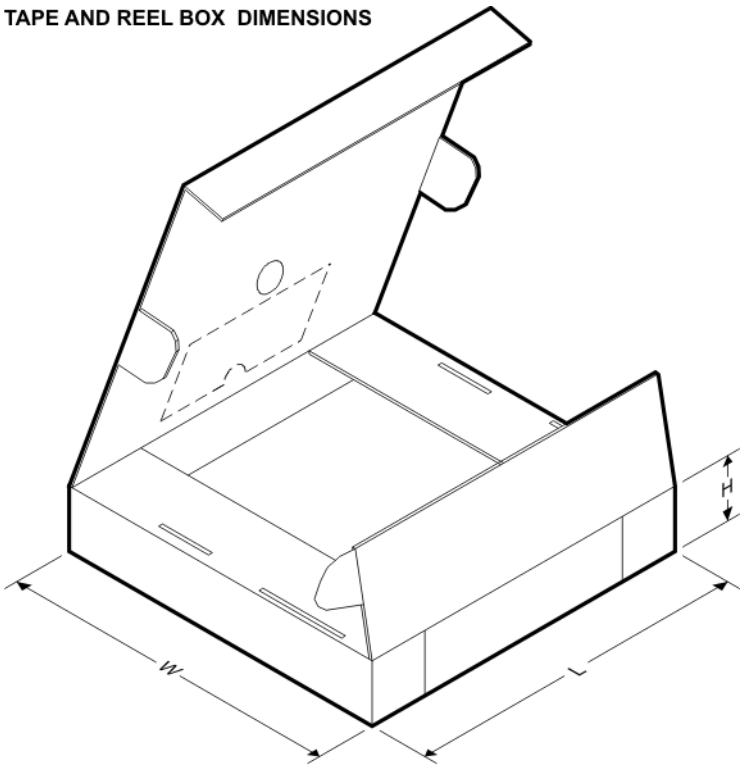
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
OPA171AQDBVRQ1	SOT-23	DBV	5	3000	180.0	8.4	3.23	3.17	1.37	4.0	8.0	Q3
OPA2171AQDGKRQ1	VSSOP	DGK	8	2500	330.0	12.4	5.3	3.4	1.4	8.0	12.0	Q1
OPA2171AQDRQ1	SOIC	D	8	2500	330.0	12.4	6.4	5.2	2.1	8.0	12.0	Q1
OPA4171AQDRQ1	SOIC	D	14	2500	330.0	16.4	6.5	9.0	2.1	8.0	16.0	Q1
OPA4171AQPWRQ1	TSSOP	PW	14	2000	330.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1

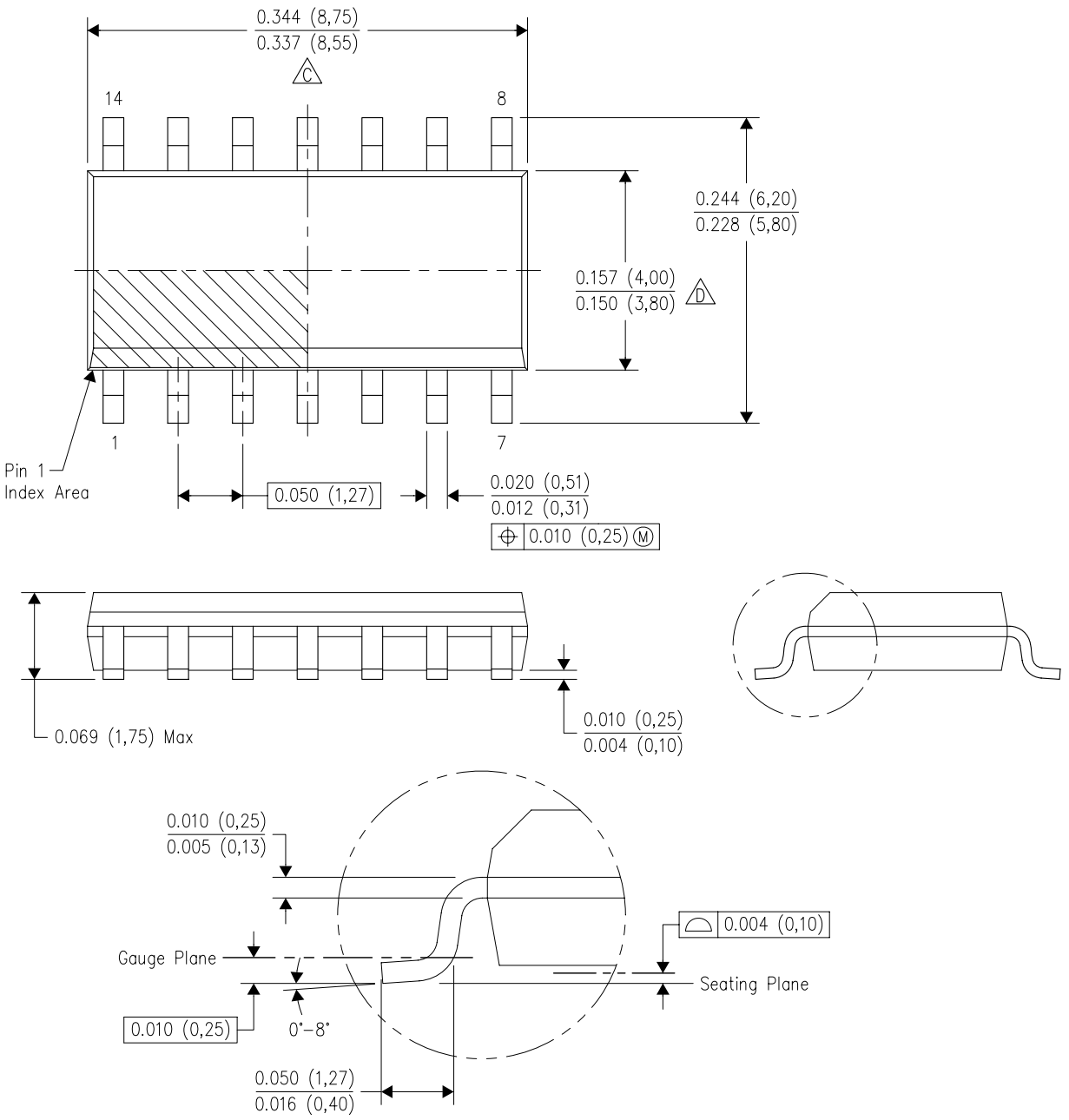
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

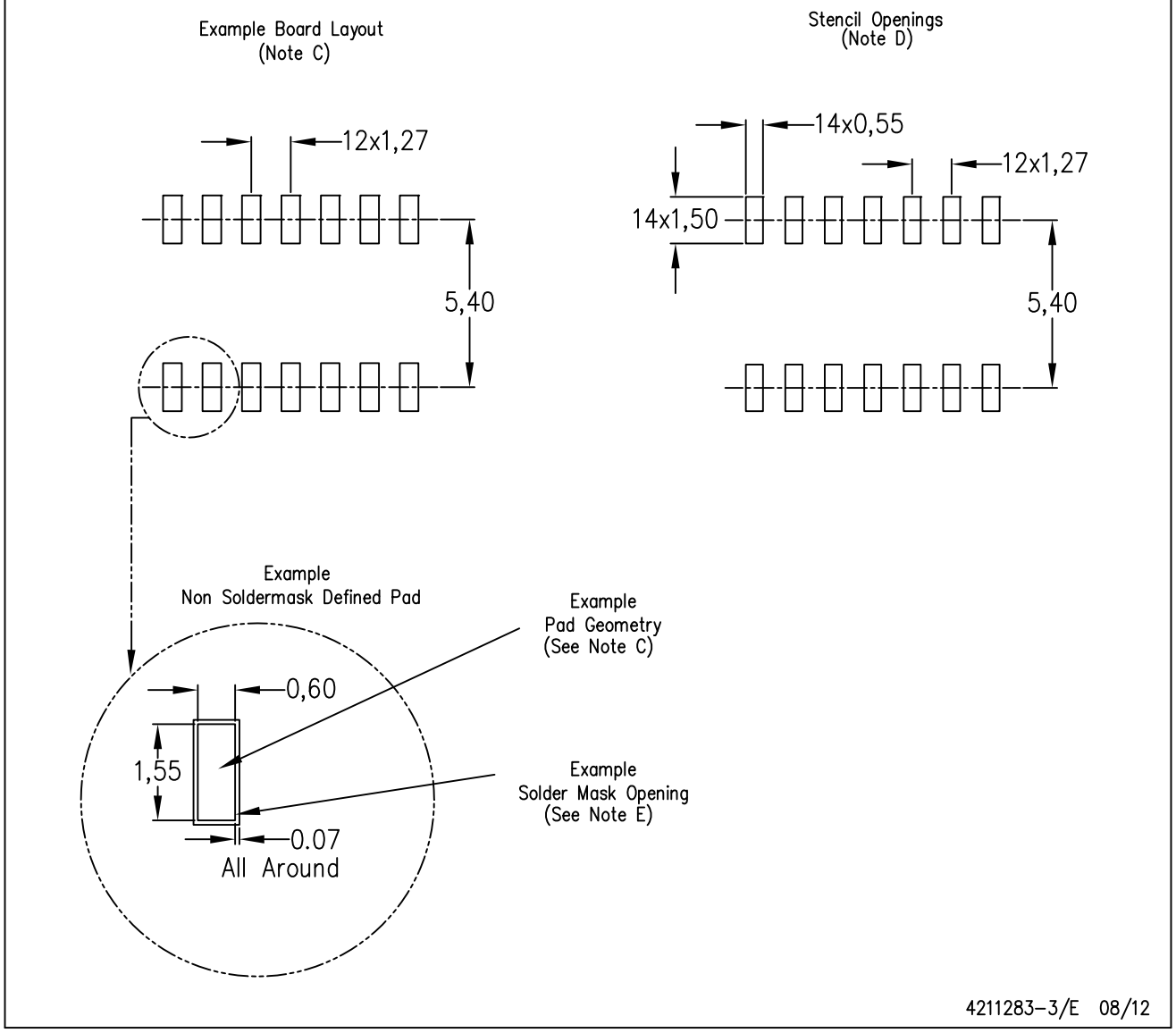
Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
OPA171AQDBVRQ1	SOT-23	DBV	5	3000	202.0	201.0	28.0
OPA2171AQDGKRQ1	VSSOP	DGK	8	2500	366.0	364.0	50.0
OPA2171AQDRQ1	SOIC	D	8	2500	367.0	367.0	35.0
OPA4171AQDRQ1	SOIC	D	14	2500	367.0	367.0	38.0
OPA4171AQPWRQ1	TSSOP	PW	14	2000	367.0	367.0	35.0

D (R–PDSO–G14) PLASTIC SMALL OUTLINE



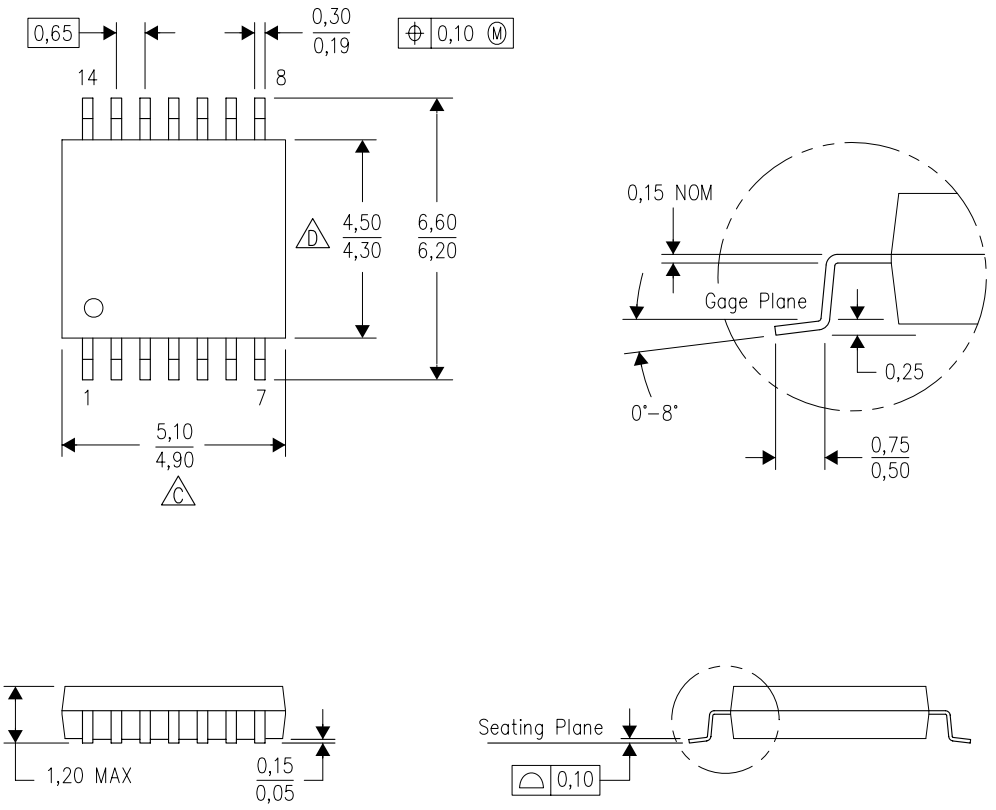
NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.006 (0,15) each side.
D. Body width does not include interlead flash. Interlead flash shall not exceed 0.017 (0,43) each side.
E. Reference JEDEC MS–012 variation AB.

D (R–PDSO–G14) PLASTIC SMALL OUTLINE



NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Publication IPC–7351 is recommended for alternate designs.
D. Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Refer to IPC–7525 for other stencil recommendations.
E. Customers should contact their board fabrication site for solder mask tolerances between and around signal pads.

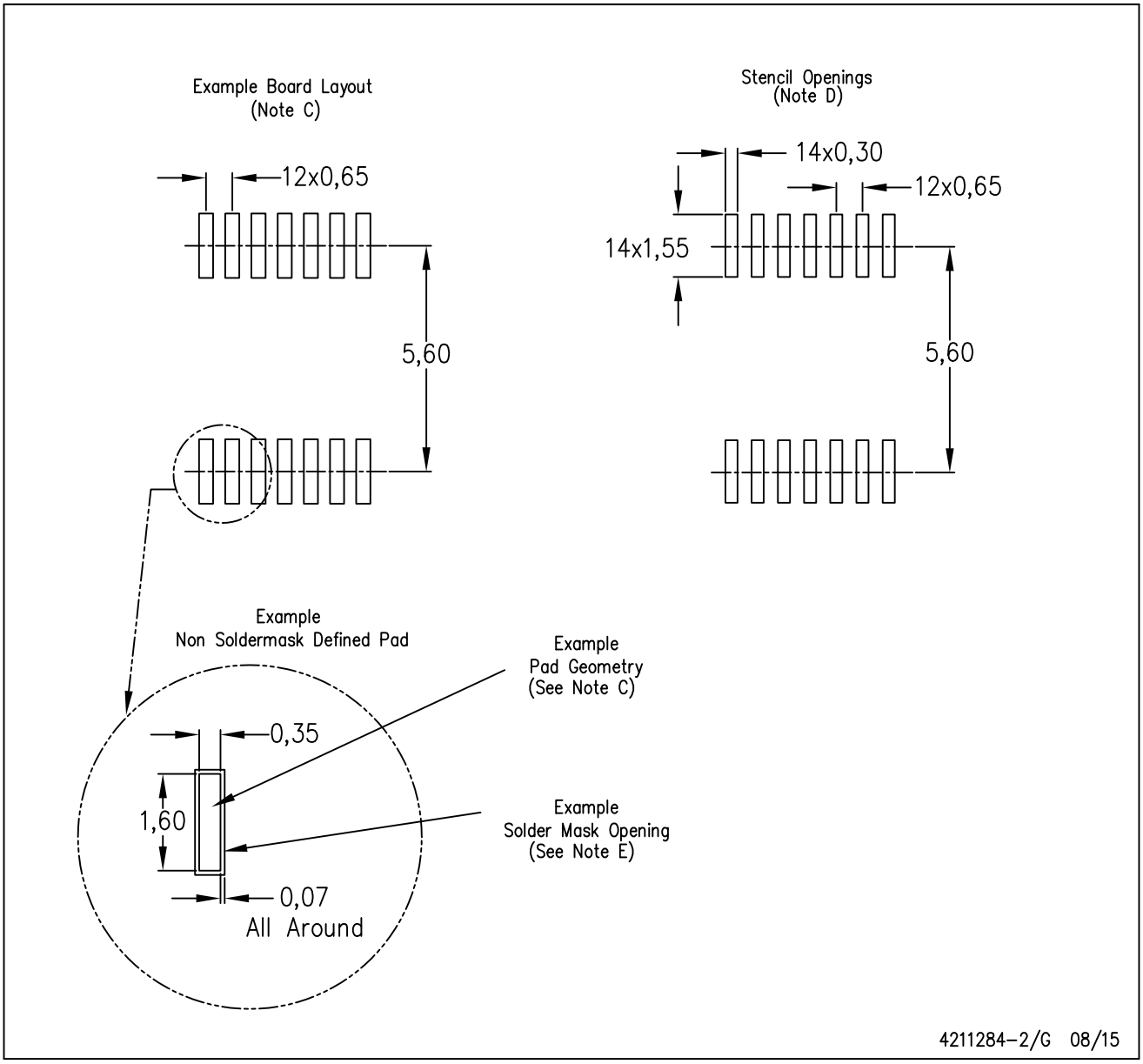
PW (R–PDSO–G14) PLASTIC SMALL OUTLINE



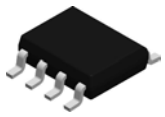
4040064–3/G 02/11

NOTES:
A. All linear dimensions are in millimeters. Dimensioning and tolerancing per ASME Y14.5M–1994.
B. This drawing is subject to change without notice.
C. Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0,15 each side.
D. Body width does not include interlead flash. Interlead flash shall not exceed 0,25 each side.
E. Falls within JEDEC MO–153

PW (R–PDSO–G14) PLASTIC SMALL OUTLINE



NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Publication IPC–7351 is recommended for alternate designs.
D. Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Refer to IPC–7525 for other stencil recommendations.
E. Customers should contact their board fabrication site for solder mask tolerances between and around signal pads.

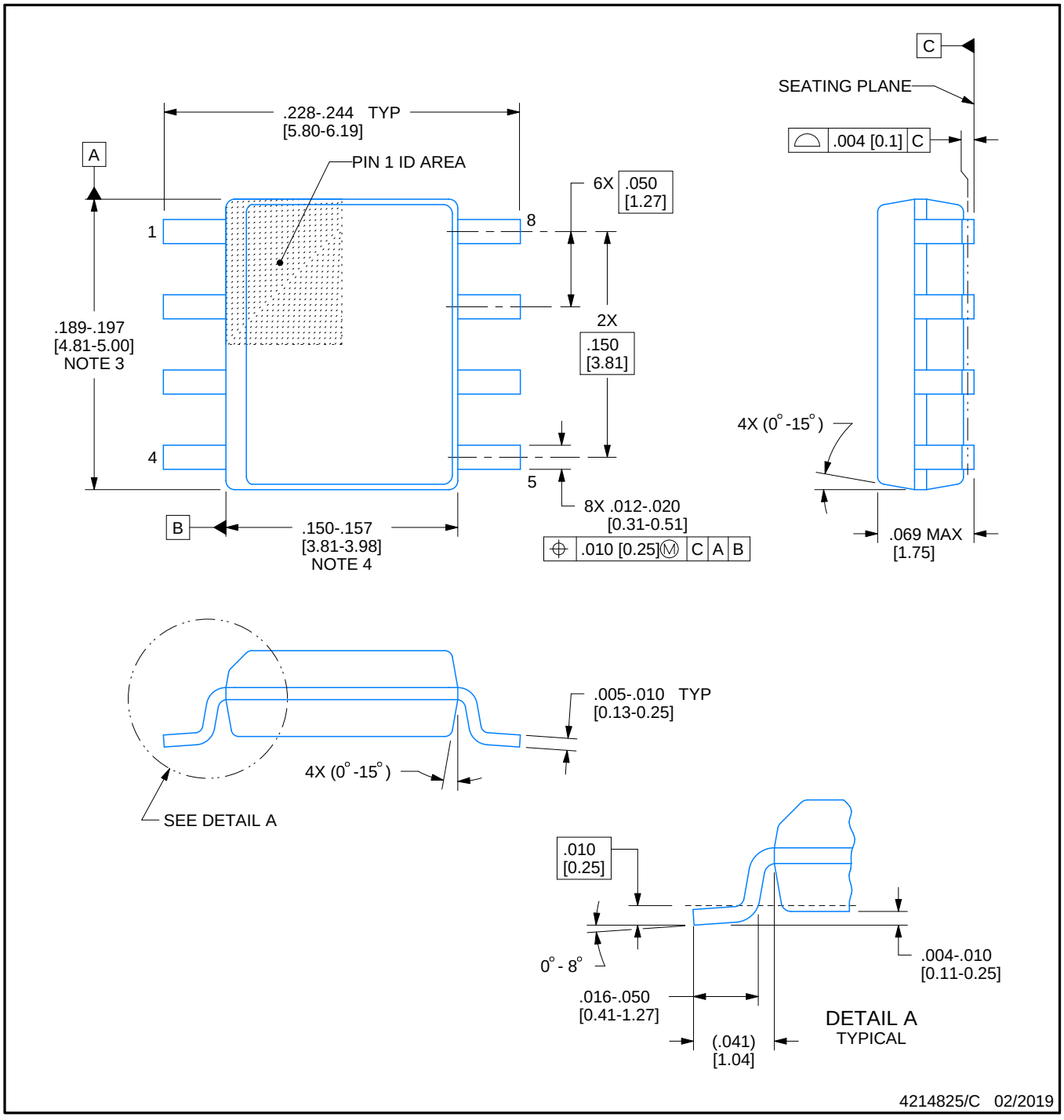


PACKAGE OUTLINE

D0008A

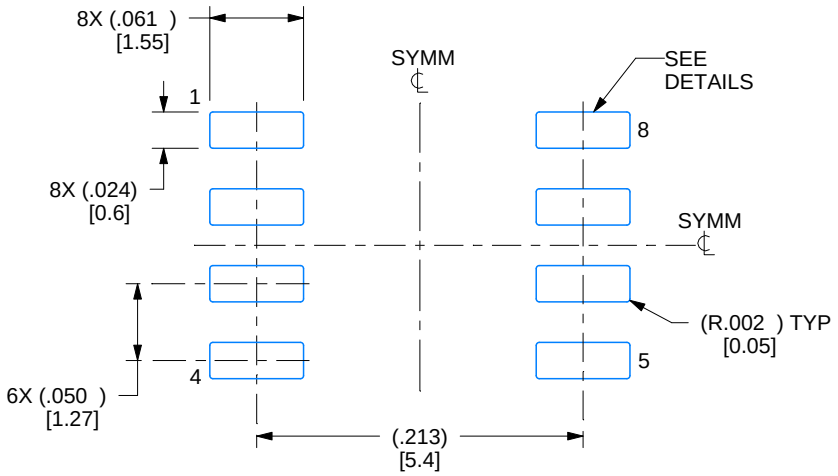
SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT

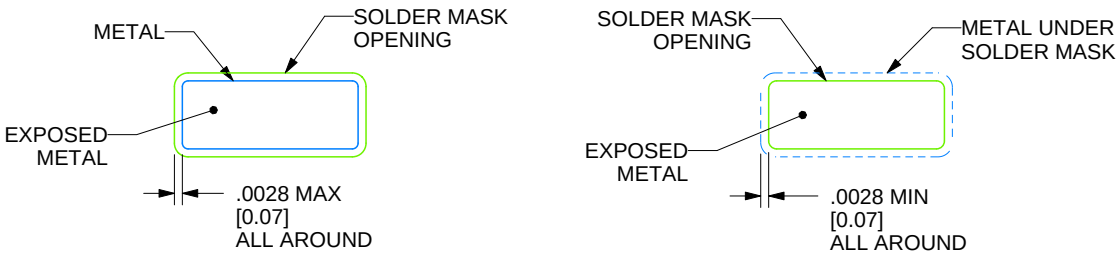


NOTES:

1. Linear dimensions are in inches [millimeters]. Dimensions in parenthesis are for reference only. Controlling dimensions are in inches. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed .006 [0.15] per side.
4. This dimension does not include interlead flash.
5. Reference JEDEC registration MS-012, variation AA.



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:8X



SOLDER MASK DETAILS

4214825/C 02/2019

NOTES: (continued)

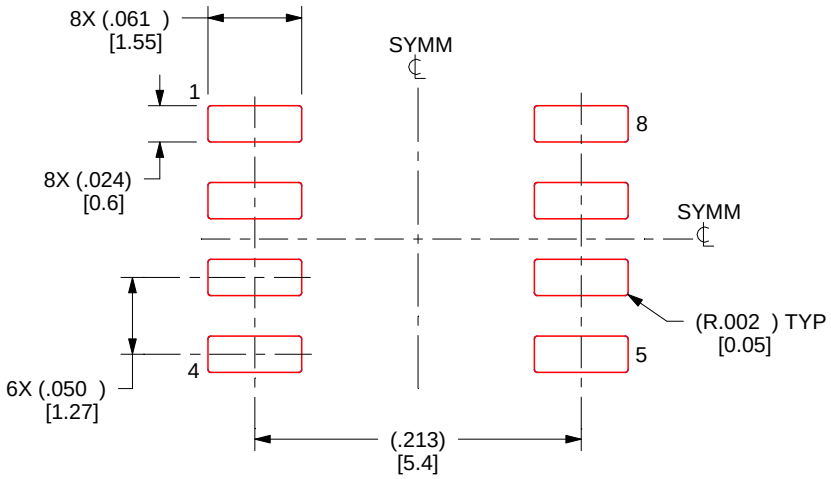
6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

D0008A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



SOLDER PASTE EXAMPLE
BASED ON .005 INCH [0.125 MM] THICK STENCIL
SCALE:8X

4214825/C 02/2019

NOTES: (continued)

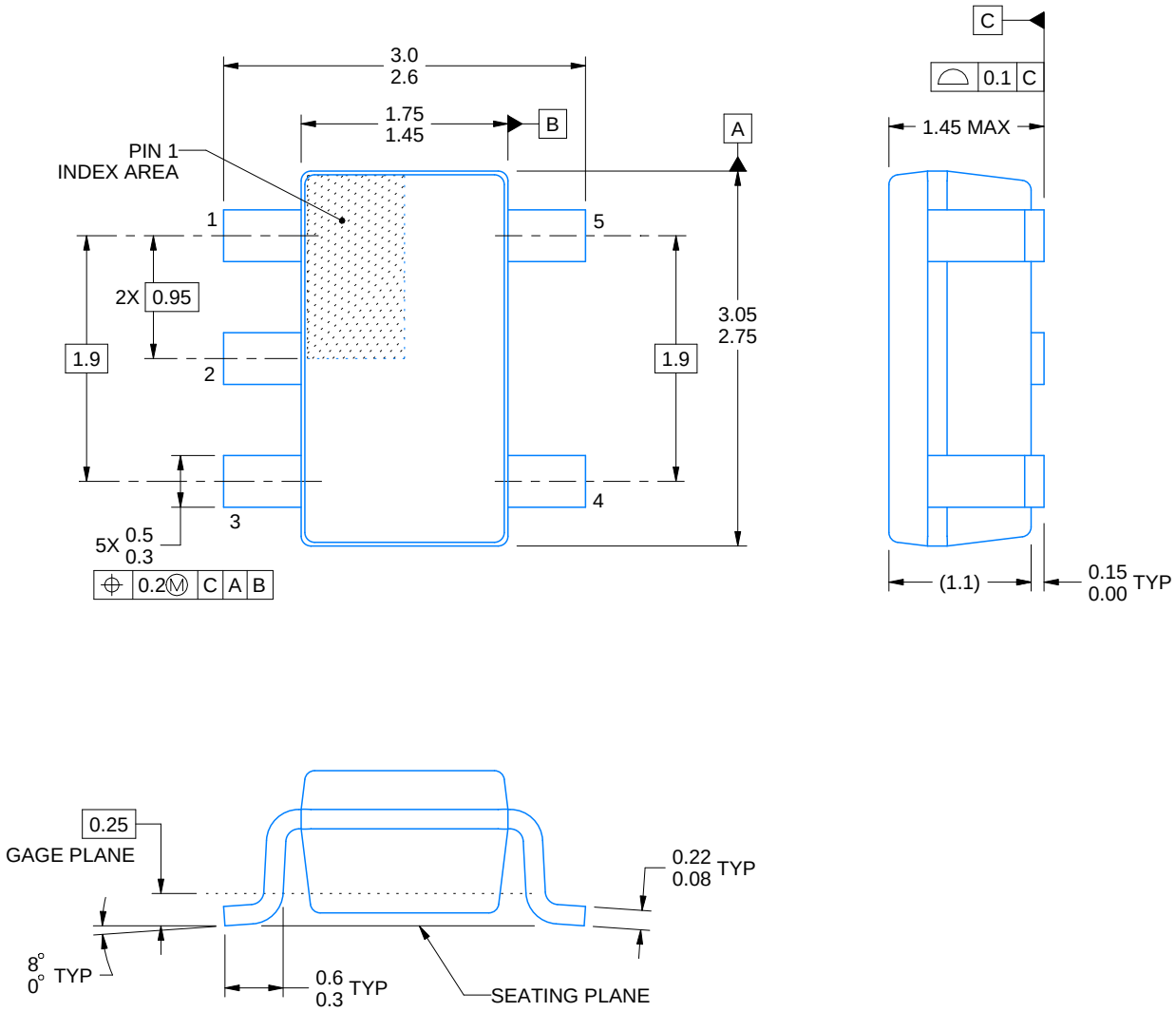
8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

PACKAGE OUTLINE

DBV0005A

SOT-23 - 1.45 mm max height

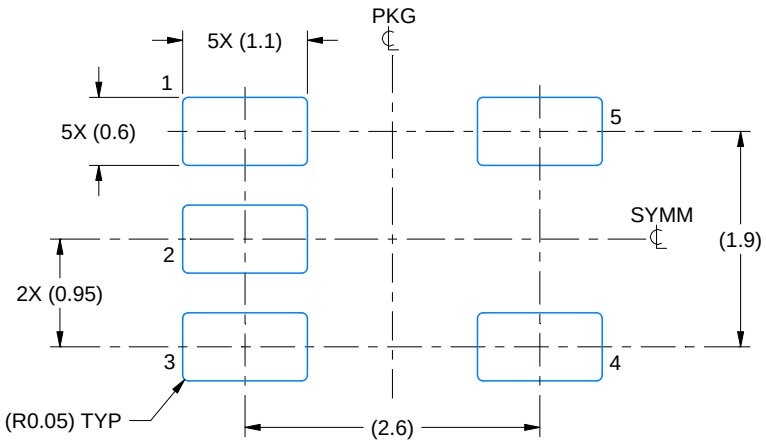
SMALL OUTLINE TRANSISTOR



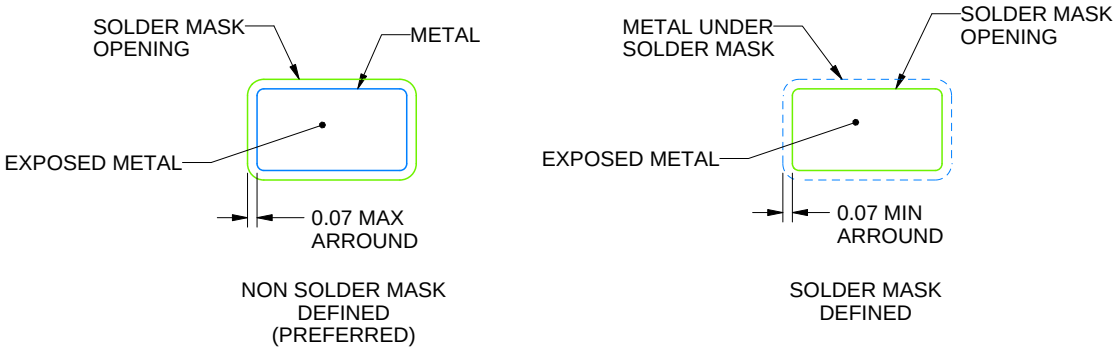
4214839/D 11/2018

NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Refernce JEDEC MO-178.
4. Body dimensions do not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:15X

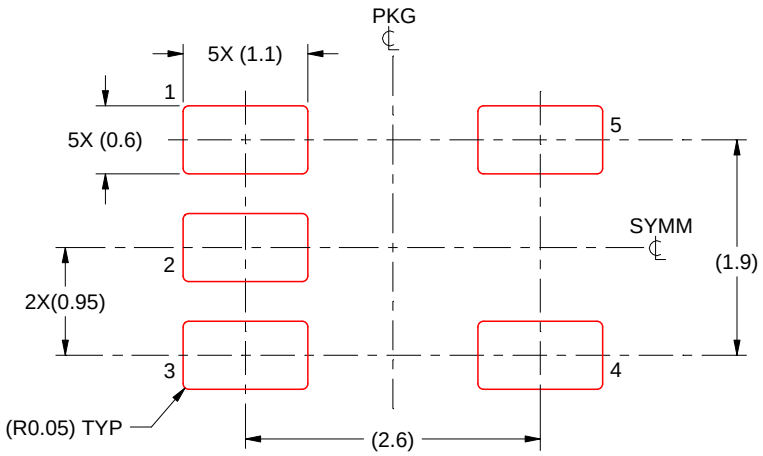


SOLDER MASK DETAILS

4214839/D 11/2018

NOTES: (continued)

5. Publication IPC-7351 may have alternate designs.
6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



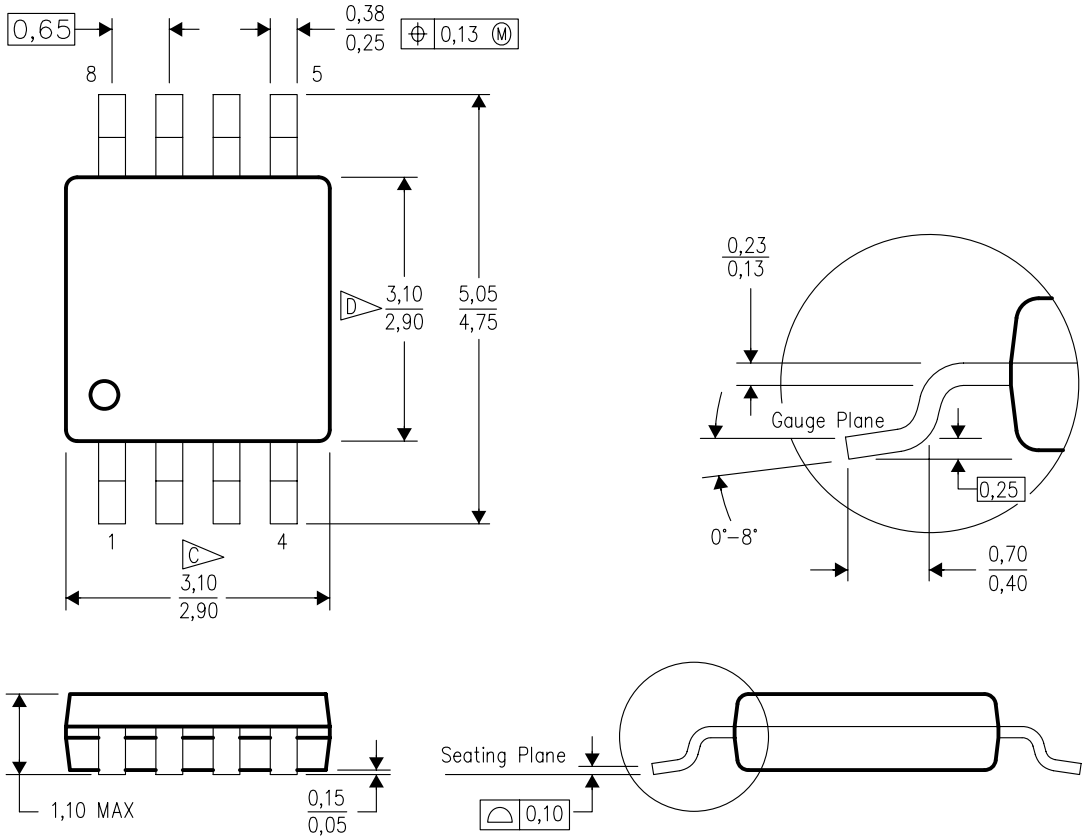
SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:15X

4214839/D 11/2018

NOTES: (continued)

7. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
8. Board assembly site may have different recommendations for stencil design.

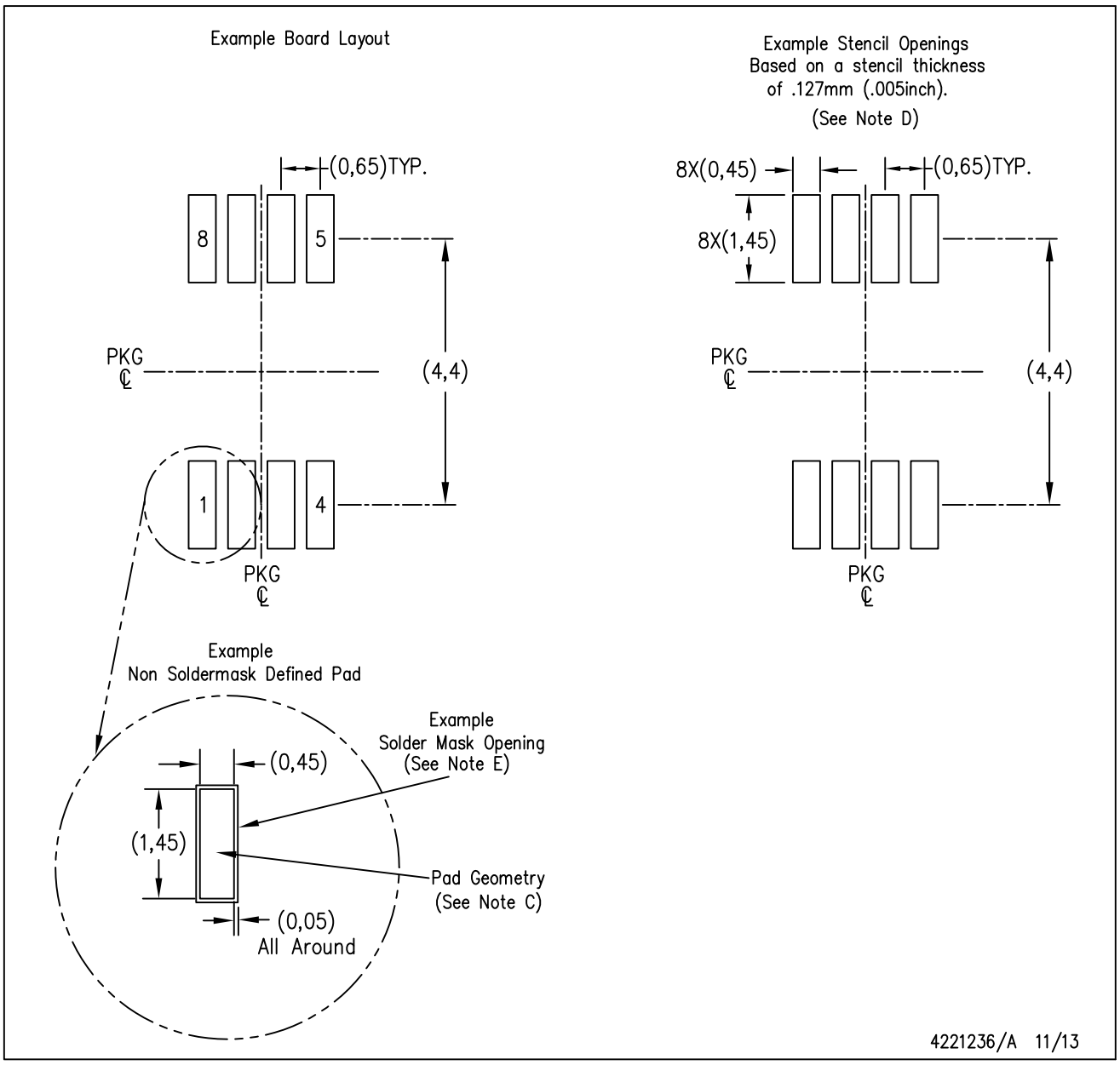
DGK (S–PDSO–G8) PLASTIC SMALL–OUTLINE PACKAGE



4073329/E 05/06

NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 per end.
D. Body width does not include interlead flash. Interlead flash shall not exceed 0.50 per side.
E. Falls within JEDEC MO–187 variation AA, except interlead flash.

DGK (S–PDSO–G8) PLASTIC SMALL OUTLINE PACKAGE



NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Publication IPC–7351 is recommended for alternate designs.
D. Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Refer to IPC–7525 for other stencil recommendations.
E. Customers should contact their board fabrication site for solder mask tolerances between and around signal pads.

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